

Connecting EXHALE to the Lower Atmosphere: Survey, Implementation, and Example Applications

1. Motivation and current state

EXHALE's lower boundary is the $\sim 1 \mu\text{bar}$ level: the input "Planet radius" is the *radius of that level* (user-guessed), the base temperature is fixed at $T_0 = T_{\text{eq}}$, the base density n_0 is an input (or pressure-anchored via `base_p_ubar`), and the gas is purely atomic H/He + trace metals at prescribed abundances. No molecules exist anywhere in the domain. The Wind-AE IC port carries a *simplified* molecular layer (μ -blend `molec_adjust` + bolometric heating/cooling, auto-shutoff when the atomic transition reaches the base).

[2026-08-15: this paragraph describes the tree as of 2026-07 and is no longer current. The coupled molecular network ($\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$, Molecular chemistry: True), H_3^+ cooling, Lyman–Werner photodissociation, the analytic lower column and the `base.inp` handoff are all implemented in `src/modules/lower_atmosphere/`. They default off, so an ordinary run is still the atomic base described here; the status table of each tier is the authoritative account.]

This memo (i) surveys how the literature connects escape models to the lower atmosphere, (ii) documents the implementation in `src/modules/lower_atmosphere/` and `src/utls/run_lower.py`, (iii) applies it to the four planets modeled with EXHALE (HD 209458 b, HD 189733 b, WASP-121 b, WASP-52 b), and (iv) prescribes how the outputs are used and connected to EXHALE runs.

2. What the literature does at the interface

Key quantitative facts:

- **Base insensitivity for strong winds** (Salz et al. 2016, §3.5–3.6): $T_{\text{base}} \pm 40\% \Rightarrow \dot{M} \pm 5\text{--}10\%$ (linear); base density $10^{14} \rightarrow 10^{12} \text{ cm}^{-3} \Rightarrow \dot{M} \lesssim 40\%$ lower. Their error budget: base density $\pm 50\%$, base $T \pm 10\%$, both dwarfed by the XUV uncertainty ($\times 3$). $\Rightarrow$ EXHALE's fixed- T_{eq} base is already defensible for hot Jupiters.
- **Where the fixed base breaks:** (i) planets near radiative equilibrium (stable thermospheres); (ii) cool/sub-Neptune planets where H_2 survives into the wind, Salz's GJ 1214 b test: H_2 10–25% throughout, H^- a significant coolant, $\dot{M} - 15\%$; Taylor et al. (2026): adding H_2 halves the He 10830 signal (14% $\rightarrow$ 7%); Koskinen et al. (2022) hot Uranus: base $q_{\text{H}_2} \approx 0.84$ and H_3^+ IR emission is the dominant radiative coolant below $\sim 2.3 R_p$.
- **What a photochemical lower model uniquely provides** (Lavvas et al. 2014; Lavvas & Arfaux 2021): the $\text{H}_2 \rightarrow \text{H}$ transition can sit exactly at $1 \mu\text{bar}$ and is temperature-sensitive there; photochemistry (OH-catalyzed H_2 destruction) dissociates H_2 even at 500–1000 K; *atomic-metal release profiles* (Na, K, Mg, Ca, Fe, Al, Si return to atomic form below $\sim 10^{-3}$ bar), the physically-motivated base abundances for EXHALE's trace metals; photoionization-dominated electron density ($\sim 10^8 \text{ cm}^{-3}$, $100\times$ Saha); radical/haze heating of $\pm 100\text{--}400$ K right at the handoff level. (2026-08-19: EXHALE carries no O/OH/ H_2O species of its own, so this chemistry reaches the code only through the handoff.)

Approach	Lower atmosphere	Hands to the wind model at $\sim 1 \mu\text{bar}$	Cost / availability
Salz et al. (2016), TPCI	none: CLOUDY solves the whole column (molecules disabled)	n/a (single domain; base at $n = 10^{14} \text{ cm}^{-3}$, T_{eq})	$\sim 3 \times 10^5$ CPU-h for 18 planets; interface not public
Koskinen et al. (2022)	<i>analytic</i> : isothermal- T_{eq} hypsometric column from 1 bar, $\mu(p)$ from chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ fits (Visscher et al. 2006)	base <i>radius</i> $r_0(1 \mu\text{bar})$, $T_0=T_{\text{eq}}$, $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$; base v from mass conservation	trivial (closed-form; all equations in the paper)
Huang et al. (2017)	<i>analytic spacer</i> : isothermal $\mu=2.3$ column, 1 bar $\rightarrow 10 \mu\text{bar}$	base radius of the atomic layer; $\text{Ly}\alpha$ <i>absorbing</i> bottom boundary (H_2 accidental resonances, $\tau=1$ at $N_{\text{H}_2} \approx 10^{14} \text{ cm}^{-2}$)	trivial
Taylor et al. (2025, 2026)	full Lavvas & Arfaux (2021) model (RC temperature + ~ 150 -species kinetics of Lavvas et al. 2014 + haze microphysics; $10^3 \rightarrow 10^{-6}$ bar)	$T(1 \mu\text{bar})$, altitude of the $1 \mu\text{bar}$ level, composition (with a <i>tolerated discontinuity</i>), K_{zz}	in-house, not public ; coupling is a manual splice, iterated $\sim 3 \times$

- **The published coupling is loose, not monolithic:** Lavvas et al. (2014) take $T(p < 1 \mu\text{bar})$ from the Koskinen thermosphere and use “species < 3 amu escape at the wind velocity” as their upper BC; Taylor et al. tolerate a composition discontinuity at $1 \mu\text{bar}$. Only TPCI runs a tight coupling at every timestep (and it disables molecules).

3. The three-tier plan

Tier 1: analytic lower column (Koskinen-2022 style), deriving the $1\text{-}\mu\text{bar}$ base radius and base $\text{H}_2/\text{H}/\text{He}$ from a closed-form column. **Tier 2:** molecular extension of EXHALE (H_2 , H_2^+ , H_3^+ , HeH^+ chemistry + H_3^+ IR cooling), required only for warm Neptunes / sub-Neptunes; **Tier 2a:** an interim EOS-only molecular-base correction. **Tier 3:** consume a real lower-atmosphere model (photochemistry + radiative-convective $T(p)$) through a file interface; the open-source VULCAN (Tsai et al. 2017, 2021) + HELIOS (Malik et al. 2017) stack is the practical replacement for the non-public Lavvas & Arfaux code, and a TPCI-style full coupling is rejected (cost, availability, and demonstrated base insensitivity). Which science case needs which tier: hot Jupiters $\rightarrow$ Tier 1 only; He 10830/ $\text{H}\alpha$ work $\rightarrow$ Tier 1; metal transmission lines $\rightarrow$ Tier 3 (atomic-metal release abundances); warm Neptunes / sub-Neptunes $\rightarrow$ Tier 2 + 1.

4. Implementation

All Fortran lives in `src/modules/lower_atmosphere/`; the Python driver in `src/utlis/`. Everything is **opt-in / default off**: with no keys and no `base.inp`, standard runs are unchanged by the lower-atmosphere code (the molecular

chemistry off-switch reproduces the atomic systems byte-for-byte; the absolute production baseline itself moved with the 2026-07-23 defaults: staged secondary ionization, He_rec_coupling, He–H charge exchange).

4.1 Tier 1: lower_column.f90

Equations. The isothermal hypsometric relation is integrated from the 1-bar level (radius $R_{1\text{bar}}$, in practice the broadband transit radius) to the base pressure $p_0 = 1 \mu\text{bar}$,

$$\frac{d \ln p}{dr} = -\frac{\mu(p, T) m_{\text{H}} g(r)}{k_{\text{B}} T}, \quad g(r) = \frac{GM_{\text{p}}}{r^2}, \quad T = T_{\text{eq}}, \quad (1)$$

with a fixed-step 4th-order Runge–Kutta in $x = \ln p$ (2000 steps; the integrand is smooth). The mean molecular weight follows the chemical-equilibrium H_2/H partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022, Eq. 11),

$$q_{\text{H}_2} \simeq \frac{1.9845 + 10^u - \sqrt{10^u (3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p_{[\text{bar}]} + 6.2645. \quad (2)$$

An implementation subtlety uncovered during validation: Eq. (2) returns the *full-mixture* volume mixing ratio $q_{\text{H}_2} = n_{\text{H}_2}/(n_{\text{H}_2} + n_{\text{H}} + n_{\text{He}})$ (a protosolar He abundance is baked into the fit), so the fraction x_2 of H *nuclei* bound in H_2 must be recovered as

$$x_2 = \frac{2 q_{\text{H}_2} (1 + f_{\text{He}})}{1 + q_{\text{H}_2}} \leq 1, \quad \mu = \frac{1 + 4 f_{\text{He}}}{(1 - x_2) + x_2/2 + f_{\text{He}}} \text{ [amu]}, \quad (3)$$

with $f_{\text{He}} = n_{\text{He}}/n_{\text{H}(\text{nuclei})}$ the elemental ratio (EXHALE’s HeH). Equation (3) was verified against the Koskinen et al. (2022) Model A base composition ($q_{\text{H}_2} = 0.84$, $q_{\text{H}} = 0.026 \Rightarrow x_2 = 0.985$); the naive H-subsystem inversion $x_2 = 2q/(1 + q)$ is *wrong* and misses the gate.

Interface. lower_column_solve(Mp, R_deep, T, fhe, p_deep, p_base, r_base, qh2_b, qh_b, qhe_b, mu_b, r_base_atomic) (cgs; pressures in bar). Because chemical equilibrium *underestimates* H dissociation at $T_{\text{eq}} \approx 1000\text{--}2000 \text{ K}$ (photochemistry ignored; Moses et al. 2011; Koskinen et al. 2013a), the routine also integrates the same column with the fully atomic $\mu = (1 + 4f_{\text{He}})/(1 + f_{\text{He}})$ and returns r_base_atomic: the two radii *bracket* the truth. The public helper q_h2_equilibrium(p, T) exposes Eq. (2) (NaN-safe, clamped to [0, 1]).

EXHALE key. Lower column: <R_1bar in R_J> in input.inp triggers a startup report: the derived $r_0(1 \mu\text{bar})$ (equilibrium + atomic bracket), the input “Planet radius” for comparison, and the base $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}/\mu$, plus a warning when the equilibrium base is strongly molecular. *Report only: nothing is overridden* (overriding is Tier 3’s job, §4.4).

Validation gate (passed). Koskinen et al. (2022) Model A (Uranus-mass planet, $M = 8.6813 \times 10^{28} \text{ g}$, $R_{1\text{bar}} = 2.5559 \times 10^9 \text{ cm}$, Sun-like star at 0.05 au, $T_{\text{eq}} = 1140 \text{ K}$, $f_{\text{He}} = 0.0793$):

	$r_0/R_{1\text{bar}}$	$q_{\text{H}_2}(\text{base})$	$q_{\text{H}}(\text{base})$
this module	1.343	0.838	0.027
Koskinen et al. (2022)	1.34	≈ 0.84	≈ 0.026

Approximations (documented in the source): isothermal column at T_{eq} ; spherical gravity (the Roche-equipotential correction matters only near RLO); chemical equilibrium only.

4.2 Tier 2a: the Molecular base key

Molecular base: True applies an *EOS-only* correction at startup: the H nuclei bound into H₂ at (1 μ bar, T_0) per Eq. (2)–(3) are removed from the base particle budget,

$$\text{ntot_bc} \rightarrow \text{ntot_bc} - \frac{x_2/2}{1 + f_{\text{He}}}, \quad (4)$$

which lowers the base pressure ($p_{\text{base}} = \text{ntot_bc} n_0 k_B T_0$) and raises the base mean molecular weight, i.e. a heavier, more compact base, the leading-order effect of a molecular base on the hydrodynamics. The species arrays remain atomic (H₂ chemistry is Tier-2 proper), so this is a *crude, clearly-flagged* approximation. [2026-08-15: Molecular chemistry: True turns this key on by itself (an atomic ntot_bc beneath a molecular base is physically inconsistent), so it need not be set separately for a Tier-2 run.] Example: HD 209458 b, $q_{\text{H}_2}(1 \mu\text{bar}, 1450 \text{ K}) = 0.831 \Rightarrow \text{ntot_bc} 1.0 \rightarrow 0.546$.

4.3 Tier 2 foundation: h3p_cooling.f90 and mol_rates.f90

H₃⁺ infrared cooling. Optically thin cooling per molecule from Miller et al. (2013, Table 5, $z(T)$ fits, archived as references/Miller_2013_JPCA_117_9770.pdf):

$$\ln E(\text{H}_3^+, T) = \sum_n C_n T^n \quad [E \text{ in W molecule}^{-1} \text{ sr}^{-1}], \quad (5)$$

piecewise over 30–300, 300–800, 800–1800, and 1800–5000 K (fit errors < 0.1% over 300–5000 K), held frozen above 5000 K (H₃⁺ is destroyed well below). The volumetric rate is

$$\Lambda_{\text{H}_3^+} = n_{\text{H}_3^+} 4\pi E(T) s_{\text{nonLTE}}(T, n_{\text{H}_2}) \times 10^7 \quad [\text{erg s}^{-1} \text{ cm}^{-3}], \quad (6)$$

where s_{nonLTE} is the departure factor of Miller et al. (2013, Table 6; their Oka & Epp detailed-balance approach), implemented as a bilinear interpolation in $(T, \log_{10} n_{\text{H}_2})$ over $T = 300\text{--}5000 \text{ K}$ and $n_{\text{H}_2} = 10^6\text{--}10^{14} \text{ cm}^{-3}$; $s \rightarrow 1$ (LTE) at high density, and the tabulated values are upper limits per the paper’s caveat on the proton-hopping rate. Both interpolators use the NaN-safe .not.(x>edge) bracketing adopted after the 2026-07-02 code review. **Validation:** the fits reproduce the paper’s Table-4 anchors at $T = 500/1000/2000/3000/5000 \text{ K}$ to <0.5%, and $s(1000 \text{ K}, 10^{10} \text{ cm}^{-3}) = 0.4955$, $s(2000 \text{ K}, 10^{12}) = 0.9499$ exactly.

Molecular reaction rates. mol_rates.f90 provides the full H₂/H₂⁺/H₃⁺/HeH⁺ thermochemical network of Koskinen et al. (2022, Table 1; transcribed and verified against the PDF page): recombination (R1–R2, Storey & Hummer 1995), Voronov (1997) collisional ionization (R3–R4), H₂⁺/H₃⁺/HeH⁺ dissociative recombination (R5–R7, R16; Larsson et al. 2008), the H₃⁺ production backbone H₂⁺ + H₂ → H₃⁺ + H (R8, 2×10^{-9}), thermal and electron-impact H₂ dissociation (R12, R14), three-body chemistry (R13, R15), the HeH⁺ chain (R17–R20), and H/He charge exchange (R21–R23). The stellar photo-rates P1–P5 (including H₂ dissociative and double photoionization) follow EXHALE’s existing PH_heat routines once H₂ cross sections are added. Lyman–Werner neutral photodissociation is not part of the source network (Koskinen et al.’s sensitivity test bounded its effect on $\dot{M}$ at $\times 1.4$); EXHALE adds it separately and opt-in, keyed on the band flux at the planet (Stellar LW flux, 2026-08-13, §9.).

Tier-2 core (implemented 2026-07-03). The coupled system `System_HeH_mol.f90` solves 7–8 unknowns per cell (H^+ , He^+ , He^{++} , optionally $He\ 2^3S$, plus the H-nucleus fractions bound in H_2 , H_2^+ , H_3^+ , HeH^+) as element-conserving steady-state balances with `hybrd1`: the atomic rows use EXHALE’s own rate arrays (so the molecule-free limit reproduces the atomic systems’ solution) and the molecular channels are the Table-1 network via `mol_rates` (R21/R22 H – He charge exchange deliberately excluded to preserve the atomic limit). The H_2 photoionization cross section (Yan et al. 1998, Eqs. 17–19; verified against their Table 7 to $<0.2\%$) enters the opacity, the P_{H_2} photoionization integral, the photoelectric heating, and the absorbed-energy normalization through optional arguments of `PH_heat_HHe`; H_3^+ IR cooling enters the `cool` array. The EOS counts each molecule as one particle, each molecular ion’s electron, and the correct masses (`calc_ne/ntot/rho`, `get_species_densities`). `f_sp` grows to 37 columns (output schema and Load-IC names extended). Key: `Molecular chemistry: True` (requires `He`; `v1` excludes trace metals; the molecular base particle count is coupled on automatically, an atomic `ntot_bc` under a molecular base is physically inconsistent). **Gates passed** (numbers re-established 2026-07-16 after the deterministic chemical-equilibrium seeding of the molecular solve; the exact gate inputs are pinned at `lower_atmosphere_figs/data_g*/input.inp`): (0) molecules-off regression, molecular chemistry off reproduces the atomic systems; (1) HD 209458 b molecules-on: a fully molecular base ($x_{H_2} \simeq 0.996$) with a sharp $H_2 \rightarrow H$ front at $r = 1.020R_p$ and an essentially atomic wind above it (the hot-Jupiter atomic base is now a *result*, not an assumption), while below the front the molecular base suppresses $He\ 2^3S$ by orders of magnitude ($He\ 2^3S + H_2$ Penning destruction, $\sim 10^4\times$ at the front, rising back to the atomic value above $\sim 1.3R_p$); (2) a hot-Uranus-like case ($0.0457 M_J$, $R_p = 0.49 R_J$, $T_{eq} = 1140\ K$, HD 209458 b orbit and spectrum): the front sits at $r = 1.156R_p$ and H_3^+ is active below it (peak $\sim 7 \times 10^4\ cm^{-3}$ at $r \approx 1.04$). The $H_2 \rightarrow H$ fronts and molecular base composition are essentially unchanged from the 2026-07-16 gates (front $1.019 \rightarrow 1.020$, $1.166 \rightarrow 1.156$), confirming the dissociation-front result is robust to the 2026-07-23 production defaults; refreshed 2026-07-23 at a shared 12000-step relaxation-snapshot convention (both HD209 runs $\log \dot{M} = 10.58$; relaxation snapshots, not flux-flat converged). Figure 1 shows the gate runs. Both gate configurations were Newton-converged on 2026-08-13 once the H_3^+ cooling entered the marching energy update; the converged solutions differ substantially from these snapshots below the $H_2 \rightarrow H$ front and are given in §7..

$He\ 2^3S + H_2$ Penning ionization. When `Include_He23S?` is also on, the metastable couples to H_2 through Penning ionization $He(2^3S) + H_2 \rightarrow He(1^1S) + H_2^+ + e^-$: the dominant $He(2^3S)$ loss toward an H_2 -dominated base (García Muñoz 2025, A&A 698, A199, Fig. 4). Its rate coefficient `penning_HeI23S_H2` (`Cool_coeff.f90`) is an analytic fit to García Muñoz Table A.5 (Cohen & Lane 1977 cross sections), $k(T) = 5.3791 \times 10^{-12} T^{0.6760} e^{-695.21/T} cm^3 s^{-1}$, reproducing the tabulated 500–10000 K points to $\leq 0.13\%$. It enters `System_HeH_mol` as an H_2 loss / H_2^+ source / $He(2^3S)$ sink and adds $(E[2^3S] - IP[H_2]) \approx 4.4\ eV$ of electron heating; the minor associative $H + HeH^+$ branch ($\sim 10\%$) is folded into the Penning channel.

The `_adv` post-process is atomic-only, so the Penning- H_2 term does not appear there. The advection-corrected reconstruction in `post_process_adv.f90` (header, lines 4–13) treats the gas as H/He + trace metals and excludes the molecular species entirely: the molecular densities are never passed in or re-solved, and its `calc_ne/calc_ntot` calls omit the `nmol` argument. Consequences for a molecular run: (i) the `_adv` n_{tot}/n_e omit the neutral- H_2 particle count

and the molecular-ion electrons; (ii) the $\text{He}(2^3\text{S})+\text{H}_2$ Penning sink cannot act in the `_adv` solve (no n_{H_2} there), so the `_adv` outputs, and the transit module that reads them, carry the *equilibrium* molecular-base triplet suppression ($39\times$ at the HD 209458 b base, $430\times$ on the hot-Uranus case) without an advection correction. This is acceptable where `_adv` is physically meaningful (the atomic/ionized escape flow above the $\text{H}_2 \rightarrow \text{H}$ front); below the front the atomic `_adv` approximation is outside its validity domain anyway. A molecular-aware `_adv` is a separate extension, naturally paired with the open “molecular advection” item in the Tier-2 remaining list.

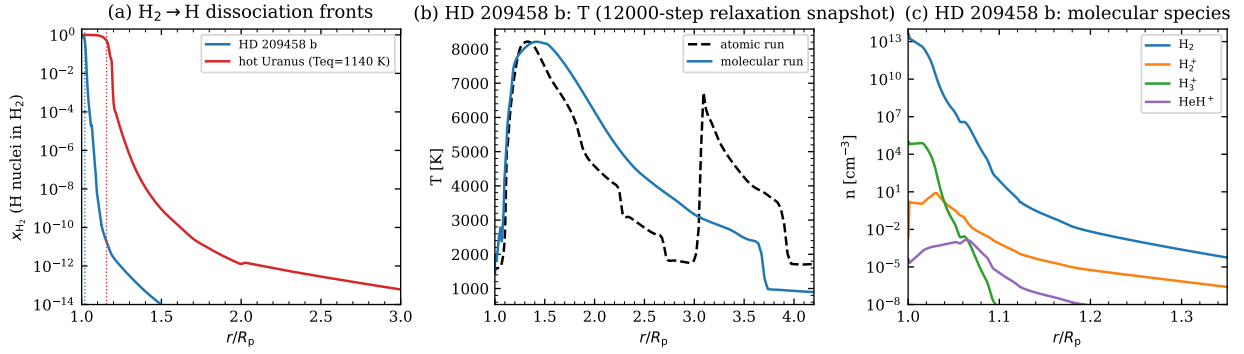


Figure 1: Tier-2 molecular gates (12000-step relaxation snapshots). (a) The H nuclei fraction bound in H_2 : on HD 209458 b the base is fully molecular and a sharp $\text{H}_2 \rightarrow \text{H}$ front sits at $r = 1.020 R_p$ (dotted); on the cooler, low-gravity hot Uranus the front sits at $1.156 R_p$. The cell-to-cell flip-flop just above the front marks the chemically bistable zone of *equilibrium* H_2 chemistry: kinetics+advection smooth it (the documented Tier-2 caveat). (b) The molecular run’s temperature agrees with the atomic run away from the (still-relaxing) front region. (c) Molecular species on HD 209458 b: H_3^+ (green) and the molecular ions are confined below the front, as expected for a hot Jupiter.

Trace metals in the same system (2026-08-13). The molecular network and the metal ionization stages are solved together in `System_HeH_mol_metals`, so a `metals.inp` may sit in a molecular run. They cannot be solved apart: in the shielded molecular base the metals are the dominant electron donors, and the free electron density is shared by both sets of rows.

Remaining Tier-2 work (open). Molecular *advection* (the local chemical-equilibrium treatment cannot reproduce Koskinen et al.’s advective replenishment of high-altitude H_2 on cool Neptunes); the dissociative/double photoionization channels (P4/P5); the 4.48 eV dissociation energy sink of the *thermal* channels R12/R14; a diatomic γ ; a molecule-aware `_adv` post-process; and the GJ 1214 b He-halving validation (Taylor et al. 2026) with a full M-dwarf setup. Lyman–Werner photodissociation (§9.) and the metals+moles merge left this list on 2026-08-13.

4.4 Tier 3: base.inp and run_lower.py

The handoff file. An optional `base.inp` in the run directory is read by `input_read` *before* the derived constants, so its overrides propagate consistently (v_0 , b_0 , grid, normalization):

key	unit	overrides	used by
T_base	K	T_0 (base temperature)	thermal BC, EOS scales
r_base	R_J	“Planet radius” ($= r_0$ of 1 μ bar)	grid, gravity, areas
HeH_base	-	He/H number ratio	composition, He diffusion
Kzz_base	cm ² /s	he_kzz	He_diffusion homopause
q_H2_base	-	q_h2_base (q_{H_2} at the base)	Molecular base particle count
p_base	bar	p_base_bar	level of the handoff; the q_{H_2} fit
<El>_H_base	-	$X_{<El>}$, hence melem_ab	metal system, EOS budget, base BC

Every override is echoed to stdout; an absent file is a strict no-op. Unknown keys and # comments are ignored, so a richer lower-atmosphere model may carry extra metadata in the same file.

Elemental reservoirs in the handoff (2026-08-26). <El>_H_base is one key per element, spelled with any of the ten element symbols of species_table (C N O Mg Si Ca Na K S Fe), e.g. O_H_base 4.90e-4; the value is the El/H *nuclei* ratio at the handoff level. It overrides metals.inp for that element and turns the metal system on by itself when it is the only nonzero abundance: thereis_metals, melem_ab and thereis_lowIP_metal are all derived after the handoff, so a handoff element behaves exactly like a metals.inp one. On a restart it also renormalizes that element’s whole loaded column onto the stated ratio, preserving the ionization split and the radial shape (load_IC.f90). Every key of base.inp now carries a *category* (provenance, EOS boundary, elemental reservoir, initial guess, boundary constraint), which is what states what the value is allowed to do to the wind; the full table is md/input_schema.md §2c and the user manual’s Tier-3 section. Neither run_lower.py nor vulcan_to_base.py writes <El>_H_base: VULCAN’s networks are H/C/N/O(/S), so the only element ratios it could hand over are its own inputs, and the atomic-metal release is outside its scope.

Tier 3 as a profile, not a level (2026-08-27). The scalars above state the lower atmosphere at one level. The key Lower atmosphere profile: <file> states it over an *interval* of pressure instead: the lower model’s own solution, tabulated deep-to-shallow and reaching above the matching level, handed over as one table. The reader is src/modules/files_I0/lower_atmosphere_profile.f90 and the schema module src/utls/lower_profile_schema.py. Two things ask for the interval. The first is eddy mixing: the homopause is where K_{zz} crosses the molecular diffusion coefficient, which a single He_Kzz cannot locate, so the profile supplies K_{zz} as a column interpolated onto the grid. The second is the elemental-flux closure of §6.3. At its matching pressure the profile sets T_0 , R_0 , p_{base} , q_{H_2} , He/H and each elemental reservoir it carries; n_tot and rho are carried but not imposed. Because it is then the single source of those quantities, a base.inp beside it may no longer state them: the EOS-boundary, elemental-reservoir and boundary-constraint keys are refused by name, only provenance comments and diagnostic keys remain allowed, and such a file must carry a # solution_id matching the profile’s or the run stops. Two producers write the format: src/utls/photochem_to_lower_profile.py (the production path, with the temperature structure either prescribed or solved by its radiative–convective climate model) and src/utls/vulcan_to_lower_profile.py (the cross-check path, VULCAN carries no climate model). Column-by-column schema: md/input_schema.md §2d; example: examples/17_lower_profile/.

The driver. `src/utls/run_lower.py <run_dir> --r1bar <R_J> [--guillot] [--tint T] [--kzz K]` reads M_p , T_{eq} , He/H from the run directory's `input.inp`, integrates the same column as §4.1 in Python, and writes `base.inp`. Two temperature models: isothermal T_{eq} (default; identical to the Fortran Tier-1 column) and the Guillot (2010) semi-grey global-average profile

$$T^4(\tau) = \frac{3T_{\text{int}}^4}{4} \left(\frac{2}{3} + \tau \right) + \frac{3T_{\text{eq}}^4}{4} \left(\frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left[\frac{\gamma}{\sqrt{3}} - \frac{1}{\gamma\sqrt{3}} \right] e^{-\gamma\tau\sqrt{3}} \right), \quad \tau = \kappa_{\text{IR}} p / g, \quad (7)$$

with standard hot-Jupiter defaults ($\kappa_{\text{IR}} = 10^{-2} \text{ cm}^2 \text{ g}^{-1}$, $\gamma = 0.4$, $T_{\text{int}} = 100 \text{ K}$). The file records the base $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$ as comments and prints a molecular-base warning when $q_{\text{H}_2} > 0.3$. The upgrade path (running VULCAN for the composition and a real RC model for $T(p)$) writes the *same* `base.inp` keys, so nothing on the EXHALE side changes.

VULCAN upgrade implemented and demonstrated (2026-07-03). The public VULCAN code (cloned to EXHALE/VULCAN/; FastChem compiled) was run to steady state for HD 189733 b (SNCHO-2025 photochemical network, 2356 steps) and converted with the new `src/utls/vulcan_to_base.py` (`.vul` $\rightarrow$ `base.inp`: photochemical $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$ at 1 μbar , hypsometric r_{base} with VULCAN's own $\mu(p)$ and $T(p)$, molecular mixing ratios as comments). **Result:** at 1 μbar the photochemical base is $q_{\text{H}_2} = 0.63$, $q_{\text{H}} = 0.23$, *photochemistry dissociates $\sim 11\times$ more H than the chemical-equilibrium column* ($q_{\text{H}} = 0.020$), directly quantifying the Tier-1 caveat (Moses 2011; Koskinen 2013a); $r_{\text{base}} = 1.168$ vs the analytic $1.174 R_J$; $T_{\text{base}} = 863 \text{ K}$ (the Moses11 $T(p)$, cooler than the isothermal $T_{\text{eq}} = 1183 \text{ K}$). The full chain VULCAN $\rightarrow$ converter $\rightarrow$ `base.inp` $\rightarrow$ EXHALE startup override was exercised end-to-end. Caveat: VULCAN's networks are H/C/N/O/(S), the atomic-metal release (Na/Mg/Ca/Fe) has no public analogue and remains with the Lavvas collaboration path; `metals.inp` stays user-supplied. Figure 2 shows the converged composition against the equilibrium H_2 track.

The photochemical q_{H_2} now reaches the solve (2026-08-10). Until then the converter recorded q_{H_2} as a comment while the Molecular base correction took its own q_{H_2} from the chemical-equilibrium fit: the very approximation the pre-step exists to replace. `base.inp` now carries two further read keys, `q_H2_base` (the photochemical H_2 volume mixing ratio at the base) and `p_base` (the level it refers to, at which the fit is evaluated so that fit and handoff always describe the same level). Both converters write them, and the base particle count is built in one place (`comp_ntot_bc` in `composition.f90`) from the photochemical value when it is present and from the fit otherwise, so a `base.inp` without the key behaves exactly as before. For HD 189733 b the VULCAN value $q_{\text{H}_2} = 0.6181$ against the fit's 0.8384 moves the subtraction from `ntot_bc` by 0.074 per nucleus ($\sim 7\%$), which the base density inherits in Base BC: pressure mode.

5. Example applications: HD and WASP planets

The approach was applied to the four planets modeled with EXHALE (`examples/13_lower_atmosphere/`; $R_{1\text{bar}} =$ broadband transit radius; T_{eq} , M_p , He/H from each run's `input.inp`).

Interpretation (per planet):

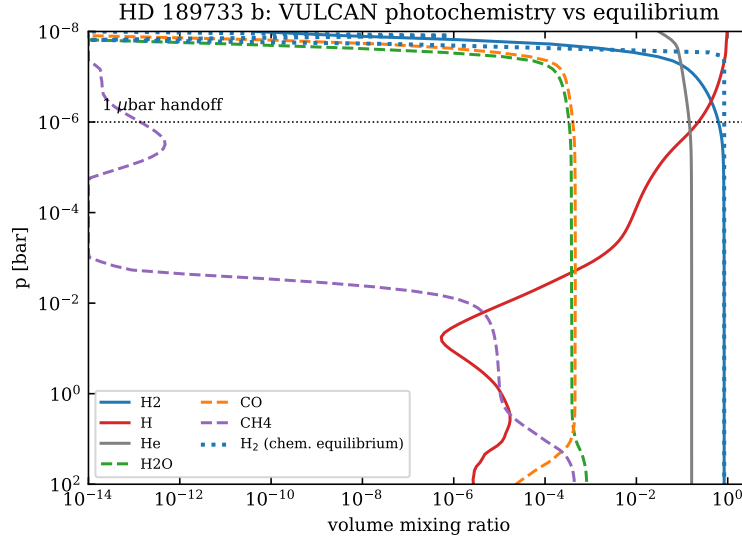


Figure 2: HD 189733 b lower/middle atmosphere from the converged VULCAN run (solid/dashed: H_2 , H , He , H_2O , CO , CH_4) compared with the chemical-equilibrium H_2 mixing ratio evaluated along the same $T(p)$ (dotted). Above $\sim 10^{-4}$ bar photochemistry dissociates H_2 far beyond equilibrium; at the $1 \mu\text{bar}$ handoff (horizontal dotted line) $q_{\text{H}} = 0.23$ versus the equilibrium 0.020 : the $\sim 11 \times$ underestimate quantified.

	M_{p}	T_{eq}	He/H	$R_{1\text{bar}}$	input R_0	$r_0(1 \mu\text{bar}) [R_{\text{J}}]$		r_0^{Gui}	q_{H_2}	μ_{base}
planet	$[M_{\text{J}}]$	$[\text{K}]$		$[R_{\text{J}}]$	$[R_{\text{J}}]$	chem.eq.	atomic	$[R_{\text{J}}]$	(base)	[amu]
HD 209458 b	0.720	1450	0.0833	1.36	1.401	1.472	1.582	1.466	0.831	2.25
HD 189733 b	1.237	1183	0.0833	1.138	1.193	1.174	1.205	1.171	0.838	2.26
WASP-121 b	1.182	2358	0.0851	1.753	2.2075	2.010	2.133	1.966	0.005	1.24
WASP-52 b	0.460	1304	0.0204	1.27	1.270	1.437	1.615	1.428	0.837	1.95

Isothermal- T_{eq} column (chem. equilibrium + fully-atomic bracket) and the Guillot variant r_0^{Gui} ($T_{\text{base}}^{\text{Gui}} = 1313, 1072, 2136, 1181 \text{ K}$ respectively). “input R_0 ” is the base radius currently used in the production runs.

- **HD 189733 b: consistent.** The production base radius ($1.193 R_{\text{J}}$) falls *inside* the derived bracket $[1.174, 1.205]$: the adopted setup is validated by the column, not merely assumed.
- **HD 209458 b: input $\sim 5\%$ low.** The production value $1.401 R_{\text{J}}$ sits just below the bracket $[1.472, 1.582]$. Given the isothermal- T_{eq} assumption this is within the method’s spread, but it suggests the $1\text{-}\mu\text{bar}$ level may lie slightly higher than assumed; the effect on $\dot{M}$ is bounded by the Salz base-sensitivity results ($\lesssim 10\%$ -level).
- **WASP-121 b: the atomic base is verified.** At $T_{\text{eq}} = 2358 \text{ K}$ the equilibrium base is already atomic ($q_{\text{H}_2} = 0.005$, $q_{\text{H}} = 0.92$, $\mu = 1.24$): for this ultra-hot Jupiter EXHALE’s atomic assumption holds *even in equilibrium*, with no appeal to photochemistry. The production base radius $2.2075 R_{\text{J}}$ (derived earlier from the Huang et al. 2023 Table-3 $\log g$ anchoring) lies $\sim 4\text{--}10\%$ above the isothermal- T_{eq} bracket $[2.010, 2.133]$: consistent with a dayside lower atmosphere hotter than the redistribution-averaged T_{eq} , which extends the column. Note the deep layers are still molecular ($q_{\text{H}_2}(1 \text{ bar}) \approx 0.83$), so μ varies from 2.25 to 1.24 along this column: the equilibrium track is

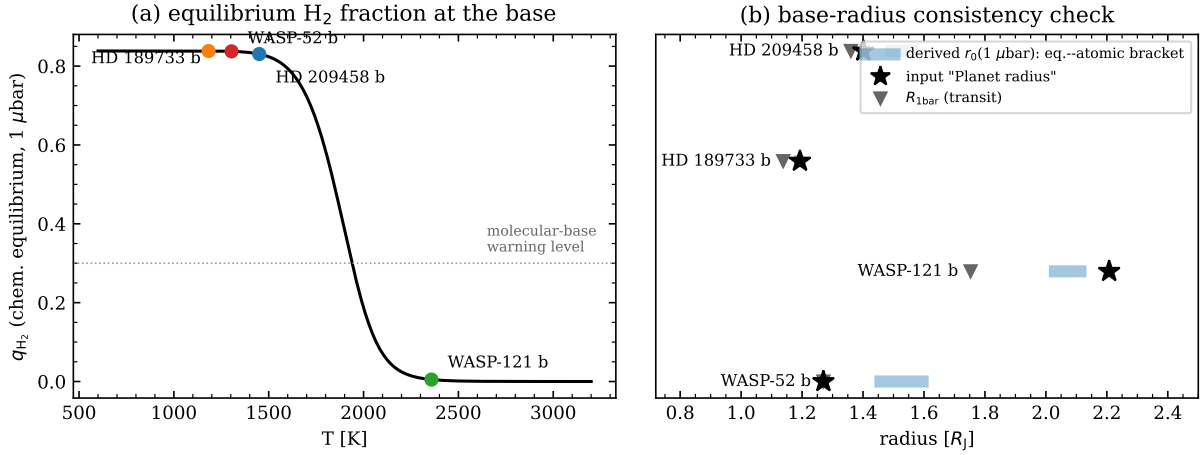


Figure 3: Tier-1 analytic column applied to the four planets. (a) The chemical-equilibrium H_2 mixing ratio at 1 μ bar (Eq. 2) as a function of temperature: the three $T_{\text{eq}} \lesssim 1500$ K planets sit on the fully molecular plateau, while WASP-121 b ($T_{\text{eq}} = 2358$ K) is atomic even in equilibrium. (b) Derived 1- μ bar base radius (equilibrium-atomic bracket, shaded) versus the production “Planet radius” (star) and the transit radius (triangle): HD 189733 b falls inside its bracket, HD 209458 b sits $\sim 5\%$ below, WASP-121 b lies just above (dayside-hot lower atmosphere), and WASP-52 b’s transit-radius shortcut puts the base 0.17–0.35 R_J too deep.

shorter than the atomic bracket here because of its heavier deep-layer μ .

- **WASP-52 b: misplaced base, impact quantified.** The production runs use the transit radius itself ($1.27 R_J$) as the 1- μ bar base, the common shortcut, but the column places that level at 1.44 – $1.62 R_J$, i.e. the base is 0.17 – $0.35 R_J$ (13–27%) too deep. *Recomputation* (the fxuv1p0_he98 configuration rerun Newton-converged with the corrected equilibrium-column base $r_0 = 1.437 R_J$, all else identical): $\log \dot{M} 11.97 \rightarrow 12.14$ ($\times 1.5$), but the He 10830 observable barely moves, line-center absorption $27.11 \rightarrow 27.56\%$ ($\times 1.02$) and equivalent width $0.713 \rightarrow 0.722 \text{ \AA}$ ($\times 1.01$). The line-forming effective radius is set by the $\tau=1$ surface of the extended wind, which stays nearly fixed in *physical* units (the larger, less-bound base raises $\dot{M}$, but the He 2^3S column at a given physical radius is almost unchanged: peak $n_{2^3S} 150.5 \rightarrow 147.6 \text{ cm}^{-3}$). Conclusion: the base misplacement biases the *mass-loss rate* at the $\times 1.5$ level but appears not to bias the He 10830 scan conclusions materially. Figure 4 shows the two line profiles. Also note this run family’s low He/H ($= 0.0204$, the “He98” abundance scan), visible in the light μ_{base} .
- **Common pattern.** All three $T_{\text{eq}} \lesssim 1500$ K planets have strongly molecular *equilibrium* bases ($q_{H_2} \approx 0.83$, the fully-molecular limit of the fit): quantifying that their atomic bases rest on *photochemical* dissociation (Moses et al. 2011; Koskinen et al. 2013a) rather than thermal, which is precisely what the Tier-3 VULCAN upgrade would supply from first principles. The Guillot column is systematically slightly more compact (cooler skin above the photosphere): a $\lesssim 0.5\%$ radius effect at fixed assumptions.

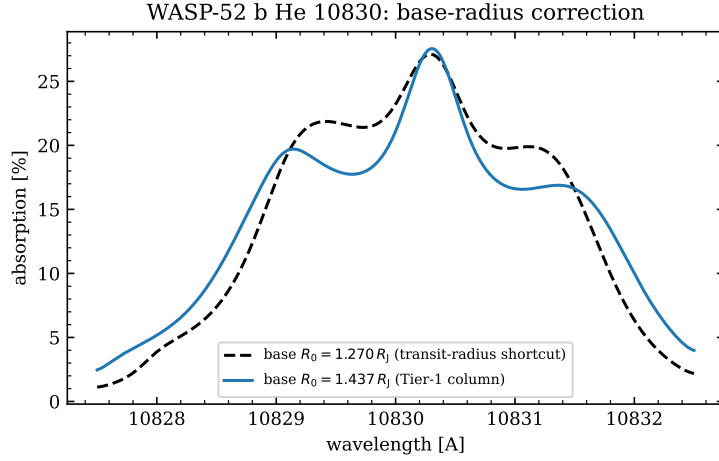


Figure 4: WASP-52 b He 10830 transmission (instrument+rotation convolved) for the old transit-radius-shortcut base ($R_0 = 1.270 R_J$, dashed) and the corrected Tier-1 column base ($R_0 = 1.437 R_J$, solid), both Newton-converged with the same binary. The line core (27.11 vs. 27.55%) and equivalent width (0.713 vs. 0.722 Å) are essentially unchanged (the $\tau=1$ line-forming surface of the extended wind is fixed in physical units) with only a mild reshaping of the shoulders; the base misplacement instead biases $\dot{M}$ ($\log \dot{M} 11.97 \rightarrow 12.14$, $\times 1.5$).

6. Using and connecting the results

6.1 Recommended workflow for each planet

1. **Column check (always, free).** Add Lower column: <transit radius> to input.inp (or run run_lower.py). Read off: does the current “Planet radius” fall inside the [chem.eq., atomic] bracket? Is $q_{\text{H}_2}(\text{base})$ small?
2. **Decide the tier.** $q_{\text{H}_2} \lesssim 0.1$ (e.g. WASP-121 b): the atomic model is verified, proceed as usual. $q_{\text{H}_2} \sim 0.8$ with $T_{\text{eq}} \sim 1000\text{--}2000$ K (typical hot Jupiters): the atomic base rests on photochemistry, proceed, but note the assumption; a genuinely cool planet (warm Neptune / sub-Neptune) needs Tier-2 physics before its He 10830 predictions are trusted.
3. **Adopt the geometry.** If the input radius lies outside the bracket (WASP-52 b), either correct “Planet radius” by hand or copy the driver’s base.inp into the run directory so EXHALE adopts r_0 , T_0 , He/H and K_{zz} automatically (echoed at startup).
4. **Run EXHALE** with the usual two-stage convergence (+ Newton finish for quantitative $\dot{M}$), then EXHALE_transit.py for the transit spectra.
5. **(Optional) iterate** once a real lower-atmosphere model is attached (§6.3).

6.2 Where each quantity connects

quantity	produced by	consumed by / effect
$r_0(1 \mu\text{bar})$	Tier 1 / base.inp	grid inner radius, gravity b_0 , irradiated area; TPM continuum radius
bracket $[r_0^{\text{eq}}, r_0^{\text{atom}}]$	Tier 1	consistency check on the input radius; systematic-error bar on r_0
T_{base}	driver (T_{eq} or Guillot)	thermal BC T_0 ; v_0 , p_0 , b_0 normalization
$q_{\text{H}_2}(\text{base})$	Eq. (2)	tier-decision switch; Molecular base correction; Tier-2 initial condition
He/H (base)	base.inp	composition; the He_diffusion reservoir value
K_{zz}	base.inp	he_kzz: homopause of the He/H (and metal) diffusive separation
$\Lambda_{\text{H}_3^+}(T, n_{\text{H}_3^+}, n_{\text{H}_2})$	h3p_cooling	(Tier 2) new term in the cool array of the energy equation
R1–R23 rates	mol_rates	(Tier 2) source terms of System_HeH_mol; $\text{H}_3^+/\text{HeH}^+$ densities for the cooling and for IR-diagnostic post-processing
metal release n_X/n_{H}	(Tier 3, VULCAN)	metals.inp values, replaces assumed solar totals for the Mg II/Ca II/Na I transmission lines

6.3 Feedback loop

One-way coupling (lower $\rightarrow$ base.inp $\rightarrow$ EXHALE, with the Taylor-style tolerated composition discontinuity) is the default.

The elemental-flux closure is implemented (2026-08-27). The two-way iteration runs on the elemental fluxes, driven by src/utis/element_flux_closure.py. The chemistry is told a trial elemental flux through its upper boundary (--trial-flux-H/--trial-flux-He of photochem_to_lower_profile.py, g s^{-1} outward positive: the Lavvas et al. 2014 prescription in flux form), the resulting profile becomes the wind's base state, and the wind's own elemental flux is measured face by face by write_element_flux_profile (binary_element_diffusion.f90),

$$F_{\text{He}} = 4\pi r^2(\rho X v + J), \quad F_{\text{H}} = 4\pi r^2(\rho(1-X)v - J),$$

and reduced to one median and one relative spread per element over a radial window. The solution is the fixed point $\Phi_{\text{El}} = F_{\text{El}}(\Phi_{\text{El}})$, reached by an under-relaxed update $\Phi^{(k+1)} = \Phi^{(k)} + \omega(F^{(k)} - \Phi^{(k)})$ with $\omega = 0.5$ halved (floor 0.125) whenever the residual fails to fall. The convergence test reads the *undamped* residual, so a small ω cannot buy a false convergence, and a residual smaller than the spread of the window it was measured on is reported UNRESOLVED rather than converged. The driver reads the overlap window first and falls back on the steady window $r \geq r_{\text{esc}}$ when the overlap is unmeasurable or too ragged, recording which window supplied the number.

Still one-way. The radiation half of the loop is not implemented. EXHALE already computes the attenuated stellar spectrum on its energy grid, so the *transmitted XUV flux at the base*, $F_{\nu}(r_0) = F_{\nu} e^{-\tau_{\nu}(r_0)}$, could be dumped and handed to the photochemical model as its top irradiation; today the adapters take a stellar flux file and a dilution instead. Taylor et al. found ~ 3 iterations sufficient for the analogous Ly α loop; the same is expected here since the wind is insensitive to base details at the $\lesssim 10\%$ level (Salz et al. 2016).

6.4 The element operator: what it conserves, and where its boundary comes from

The closure above rests on the same operator that carries the He/H and metal separation through the wind, `binary_element_diffusion.f90`. Three statements about it were settled on 2026-09-10 and belong beside the flux numbers, because two of them change what those numbers mean.

The projection now conserves the mixture mass. After each element update the species vector is projected back onto the new element totals by `project_elements`. It used to take the mixture mass as $m_{\text{sum}} = m_1 n_{\text{H}} + m_{\text{He}} n_{\text{He}}$ with m_1 the *reservoir* mass per hydrogen nucleus, i.e. the trace metals at their `metals.inp` abundances, while the cell carries its metals at whatever the trace transport left it. The difference was written into hydrogen at every call. MEASURED at one cell of the certification column whose metal/H was 0.024401 against the reservoir's 0.028227, $\sum_i m_i f_i$ went $1.0000000 \rightarrow 1.0000636$ in a single step: mass created. A second leak of the same kind: the trace-metal transport wrote metal densities with no counter-flux in the hydrogen they diffuse through (3.0×10^{-3} on the suite column with the metals frozen).

`mixture_mass_split` now reads the mass the cell's species actually carry, component 1 is scaled as a whole, and one projection after the metal loop puts the metals' mass change back onto hydrogen. Mass closure of the certification column after the relaxation went $3.37 \times 10^{-3} \rightarrow 2.3 \times 10^{-15}$, and of the handed-back atomic candidate $6.44 \times 10^{-3} \rightarrow 1.94 \times 10^{-9}$, the file's own precision. A run with metals off is unaffected by construction ($m_1 = m_{\text{H}} = 1$ exactly). The counter-flux is closed on hydrogen but not on momentum or internal energy; that is a trace approximation and is named as one at the routine.

This is not a cosmetic repair for anyone using the coupling: the `lower_profile` regression case, the one that hands the lower atmosphere over as a profile, had a stored reference 3.6×10^{-4} out of closure (O/H 4.61×10^{-4} against the reservoir's 4.90×10^{-4}). A 1500-step cold start goes $6.77 \times 10^{-3} \rightarrow 3.14 \times 10^{-9}$ in closure, with T at $1.6 R_{\text{p}}$ moving 1.3% and H 6.3%. Its reference was re-snapshotted at the 2026-09-10 gate.

The boundary of the residual is the caller's, deliberately. The two relaxation solvers (`element_diffusion_step`, `solve_trace_element_in_hydrogen`) refill the upper ghosts of their own working arrays before every pass, while `element_transport_residual`, which measures the same balance on a state, reads the ghosts the caller left. Making the residual pose its own zero-gradient top was tried and reverted. It is physically the right condition and it changes the residual by one element's outermost row in its twelfth digit and nothing else (the production callers already hand the operator ghosts equal to the cell to one to four units in the last place), but the atomic element rows of the stationary solve do not survive a last-bit change of those ghosts, whatever makes it: a control that merely multiplies the same three ghost quantities by $1 + 2 \times 10^{-16}$ and poses no boundary at all moves them just as far. The amplifier is the Krylov leg at its one-digit tolerance, which crosses that tolerance on a different vector and sends the acceptance test down a different branch a few iterations later. The asymmetry therefore stays in the tree, documented at the routine, which records the size of the caller dependence (1.19×10^{-2} of the row's scale on a stale caller) rather than assuming it away.

The open item: the layer does not conserve the element flux. The window reduction of the flux closure above is also the honest statement of how well the operator conserves what it transports. MEASURED on the atomic candidate

state, as the relative radial spread of F_{El} about its median:

window	F_{He} spread	F_{H} spread	face mass-flux spread
layer, $r < 1.10$	9.79	9.09	9.25
wind, $r \geq 1.20$	2.71×10^{-2}	1.36×10^{-2}	3.99×10^{-3}
escape, $r \geq 2.0$	2.01×10^{-3}	2.14×10^{-3}	1.55×10^{-3}

The mass-flux spread the same state is usually quoted with, 9.8×10^{-7} , is a *cell-centered* number and says nothing about these: the element fluxes are three to four decades worse in the same state. The projection fix buys a factor four in the layer ($39.3 \rightarrow 9.79$ for helium) and nothing in the wind, which is what a mass-closure repair should do, since the leak was largest where the composition changes most. A spread of 9.8 means the layer's element fluxes are not conserved at all, and the standing base wave is the suspect. The consequence has been taken: the certification gates the element and carrier rows only at $r \geq 1.20$, and the layer's rows are reported and do not gate until this number is repaired; certification.f90 states that reason at the declaration of its two regime radii. Numbers here are READ from the certification tolerance anchoring measurement, not re-measured here.

6.5 Consistency cautions

- Do not double-count composition: if `base.inp` sets He/H from a lower model, the `metals.inp` abundances should be the same model's *atomic release* values, not solar totals.
- `base.inp` overrides `win` over `input.inp` for T_0 , R_0 , He/H (echoed); remove the file to return to the manual values.
- The Tier-1 column is isothermal and spherical: near-RLO planets (WASP-121 b) need the Roche correction before the bracket is taken quantitatively, and dayside-hot lower atmospheres extend beyond the T_{eq} column (both visible in the WASP-121 b example above).
- The `Molecular` base key changes only the EOS budget; do not combine it with quantitative He 10830 claims for genuinely cool planets: that is Tier-2 physics.

7. Converged Tier-2 solutions (2026-08-13)

Every Tier-2 number quoted above comes from a 12000-step relaxation snapshot: until 2026-08-13 no molecular configuration in the tree had ever reached a steady state (examples/15_molecular ran 135694 steps to $du = 2.9 \times 10^{-2}$). The first Newton-converged molecular solutions were obtained on the hot-Uranus gate (lower_atmosphere_figs/data_g2/input.inp, the configuration the `mol_base_handoff` regression case runs, $q_{\text{H}_2}^{\text{base}} = 0.75$) once the H_3^+ infrared cooling was moved inside `eval_cool` (docs/Update_EXHALE.pdf §54), so that the marching temperature update and the steady residual balance the same cooling function.

Recipe. Three keys, and they are the whole change:

Solver: Newton 5.0e-2 Resid tol: 1.0e-5 Max steps: 150000

Solver: Newton 5.0e-2 raises the hand-off threshold, because the du descent of a molecular run is *not monotonic* (on the hot-Uranus gate it reaches 8.8×10^{-2} near step 11500 and bounces back to 7×10^{-1} before resuming), so a run can spend its whole step budget above the 10^{-2} default. Max steps: 150000 covers the marching warm-up. Marching alone can *never* reach the molecular layer: its H_3^+ cooling time is 1.3×10^7 s at $r = 1.02$ against a CFL-limited step of 0.30 s, i.e. 4×10^7 steps per cooling time. It is the Newton solve that reaches it.

The default residual target is too loose for a molecular run. $\|R\|$ is the maximum relative residual over the wind, and in a molecular run it is set by the first two or three cells above the base, where the energy residual is $\sim 10^3$ times the value anywhere else. Stopping at the default $\|R\| < 10^{-3}$ therefore leaves the molecular layer still cooling: its cell-by-cell energy residual is then $0.7 \times$ the local H_3^+ cooling rate, with the same sign in 173 of 180 cells. Re-solving the same state with Resid tol: 1.0e-5 moves the layer again (by 60–70% in T), and only there does the solution become a fixed point of the procedure: a third solve reproduces T to $\leq 0.14\%$ and the $H_2 \rightarrow H$ front to five digits. **A molecular run needs Resid tol: 1.0e-5.**

hot-Uranus gate	metals off snapshot $\rightarrow$ converged	metals on (solar C/N/O/Mg/Ca/Na/Fe) snapshot $\rightarrow$ converged
residual norm at exit	6.7×10^{-7}	1.4×10^{-6}
$H_2 \rightarrow H$ front ($x_{H_2} = 0.5$)	1.16037 $\rightarrow$ 1.03581	1.16244 $\rightarrow$ 1.07974
base T (cell 1, pinned by the BC)	1213.42 $\rightarrow$ 1212.86 K	1213.46 $\rightarrow$ 1213.45 K
H_3^+ peak [cm^{-3}]	$7.03 \times 10^4 @ 1.0419 \rightarrow 3.75 \times 10^5 @ 1.0148$	$52.1 @ 1.1355 \rightarrow 1.01 \times 10^3 @ 1.0789$
T at $r = 1.02$	1304.9 $\rightarrow$ 299.0 K	1312.8 $\rightarrow$ 928.5 K
T at $r = 1.08$	1302.6 $\rightarrow$ 1075.5 K	1319.8 $\rightarrow$ 208.9 K
$\log_{10} \dot{M}$ [g/s], spherical	(not flat) $\rightarrow$ 10.542	(not flat) $\rightarrow$ 10.599
mass-flux spread, $r > 1.2$	$2.3 \times 10^2 - 3.8 \times 10^3 \rightarrow 2.4 \times 10^{-3}$	idem

The snapshot $\dot{M}$ is quoted only because the code prints it: its mass flux varies by a factor $2 \times 10^2 - 4 \times 10^3$ across the wind, so no single number represents it.

What the converged solution says physically. The molecular layer collapses to 190–300 K (far below $T_{\text{eq}} = 1140$ K), and stops there because the coolant switches itself off: the H_3^+ cooling time at $r = 1.02$ rises from 1.3×10^7 s in the snapshot to 1.3×10^{10} s in the converged state. With metals the collapse is instead accompanied by the saturated ground-term fine-structure lines (C I 609/370 μm at the base, [O I] 63/145 μm above it: 89% and 91% of the local cooling at $r = 1.002$ and 1.040), and H_3^+ never exceeds 2.3%. Either way the layer is cooled by lines treated as *optically thin*, with no thermal-infrared heating from below and no radiative equilibrium: the base temperature is pinned in one cell by the boundary condition and nothing holds the column above it. A real H_2 atmosphere at these column densities is optically thick in exactly these lines and sits near the radiative–convective profile. The converged Tier-2 thermal structure should therefore be read as **the steady state of the model as written, and as a quantitative statement that the model was**

missing an infrared escape probability below the $\text{H}_2 \rightarrow \text{H}$ front, not as a prediction of the temperature of a warm Neptune's lower thermosphere. §8. supplies part of what is missing and measures how far it goes. The wind above the front is much less affected: $\dot{M}$ moves 0.085 dex (metals off) and 0.043 dex (metals on) between the 10^{-3} and 10^{-5} solutions, because it is launched above the collapsed layer.

HD 209458 b: the same recipe on a real planet. The second Newton-grade molecular solution is examples/15_molecular: the metals-off HD 209458 b configuration of Gate 1, the one that had run 135694 steps without ever converging. The three keys above are the whole change: no warm restart was needed, the run went from the default cold IC to $\text{info} = 0$ in one pass, 10 min 25 s on 8 threads. PLM $\rightarrow$ WENO3 at step 13299, secondary ionization flipped in at step 23837, JFNK engaged at step 25837 and took $\|R\|$ from 4.031×10^{-2} to **9.542×10^{-6}** in 272 iterations. Here too the du descent is not monotonic. The residual is dominated by the same place: the worst cell is $j = 1-3$, $r \approx 1.000-1.001$, for 267 of the 272 iterations, split about evenly between the momentum and the energy row.

HD 209458 b, metals off	12000-step snapshot	converged ($\ R\ = 9.5 \times 10^{-6}$)
$\text{H}_2 \rightarrow \text{H}$ front ($x_{\text{H}_2} = 0.5$)	1.01975	1.00891
base T (cell 1, pinned by the BC)	1466.55 K	1466.36 K
T at $r = 1.005/1.02/1.08$	1909 / 1916 / 2855 K	401 / 2206 / 7574 K
coldest cell below $r = 1.3$	1463.6 K @ 1.0002	397.3 K @ 1.0056
H_3^+ peak	$9.57 \times 10^4 \text{ cm}^{-3}$ @ 1.0004	$5.18 \times 10^5 \text{ cm}^{-3}$ @ 1.0017
x_{H_2} at the base	0.9964	0.9963
$\log_{10} \dot{M}$ [g/s], spherical	10.873	10.532
mass-flux spread over $r > 1.05$	1.23	4.2×10^{-3}

The solution is a fixed point of the procedure: re-solving from it with the same tolerance returns $\text{info} = 0$ at $\|R\| = 9.0 \times 10^{-6}$ and reproduces the front to 1.00879 (0.01%), $\dot{M}$ to 0.002 dex and the H_3^+ peak to 0.4%; only the coldest cell of the collapsed layer still moves (397 $\rightarrow$ 353 K). Note the reload does not preserve the residual: Load IC? True re-derives the ionization state from the file, so the second solve starts at $\|R\| = 8.7 \times 10^{-4}$. The physical reading is the hot-Uranus reading, on a hot Jupiter: the molecular layer between the base and the front collapses to ~ 400 K, H_3^+ carries **100%** of the radiative cooling from the base out to $r \approx 1.005$ (10^4 times the local photoheating at the base), and the same optically thin treatment applies. The front moves inward by $0.011 R_p$ and $\dot{M}$ falls by 0.34 dex relative to the snapshot, so for HD 209458 b the relaxation-snapshot $\dot{M}$ is *not* a substitute for the converged one.

With metals the same planet does not converge. examples/16_molecular_metals (the identical configuration plus solar C/N/O) under the same three keys reaches the hand-off (PLM $\rightarrow$ WENO3 at step 15263, secondary ionization at 61158), and the JFNK then stalls: $\|R\|$ falls from 2.338×10^{-2} to 3.6×10^{-4} over 281 iterations and the line search finds no descent step for 12 consecutive iterations ($\text{info} = 2$, worst cell $j = 93$, $r = 1.021$, momentum), after which the run reverts to marching. This is the reverse of the hot-Uranus gate, where the metals-on case converged as readily as the metals-off one, and it is the base-adjacent momentum stall recorded as open item (A), displaced outward to the front. 16_molecular_metals therefore carries the plain Solver: Newton line and has no converged solution.

8. The infrared field of the lower atmosphere (Base IR field)

The converged Tier-2 solutions above left one question open: why does the molecular layer sit at 190–300 K, and is the optically thin line cooling the cause? The measurement below says the suspect is the right one for the metals-off case and the *wrong* one for the metals-on case, and the closure follows the measurement.

What the lines actually see. Line-center optical depths of the eight ground-term fine-structure lines, from the converged metals-on solution of the hot-Uranus gate, with the level populations and Doppler widths the cooling module itself uses. τ_{up} is the column from the cell to the top of the domain (the one the escape probability was built from), τ_{dn} the column to the bottom.

r/R_p	T [K]	$\tau_{\text{up}}(\text{CI } 609)$	τ_{dn}	β	$\tau_{\text{up}}([\text{O I}] 63)$	τ_{dn}	β
0.9998	1213	0.088	1.7×10^{-4}	0.904	2.01	4.0×10^{-3}	0.211
1.0102	1033	0.069	0.019	0.924	1.57	0.44	0.265
1.0302	820	0.040	0.048	0.955	0.89	1.1	0.421
1.0501	573	0.020	0.068	0.978	0.43	1.6	0.633
1.0749	289	0.005	0.083	0.995	0.093	1.9	0.898
1.0832	188	0.001	0.086	0.998	0.028	2.0	0.968

So **escape was never the problem**: outward the collapsed layer is thin ($\beta \geq 0.90$ for CI, ≥ 0.63 for [O I] above $r = 1.05$), and trapping can only reduce the cooling by a factor of a few at the base. What the treatment left out is the *other* direction. The gas below the base is not vacuum. Taking the base cell opacity and one pressure scale height of an isothermal exponential reservoir (base pressure 9.0 μbar):

line	τ per scale height at the base	pressure where $\tau = 1$	τ at 1 bar
[O I] 63 μm	1.08	17 μbar	1.2×10^5
[O I] 145 μm	0.32	37 μbar	3.6×10^4
CI 370 μm	0.092	106 μbar	1.0×10^4
CI 609 μm	0.046	205 μbar	5.1×10^3
C II 158, N II 205/122, [O I] 44 μm	$\leq 1.2 \times 10^{-10}$	-	$\leq 3 \times 10^{-2}$

Every line that carries the cooling is black downward within one to three scale heights below the model base. The layer therefore faces a blackbody, not a void, and the ratio of the incident mean intensity to the line source function, $\bar{J}/S = \frac{1}{2}B_\nu(T_{\text{base}})/B_\nu(T)$, crosses 1 wherever T drops below ~ 610 K: at the coldest cell it is 3.4 (CI 609 μm) to 5.7 ([O I] 63 μm). Those lines were being made to cool a gas that they should have been heating.

But that is not what makes the metals-on layer cold. Two measurements say the fine-structure lines are not the cause of the 190–300 K there.

- **The specific entropy rises monotonically outward through the whole collapsed layer**, p/ρ^γ going from 1.00 at the base to 1.67 at $r = 1.05$ and 5.69 at $r = 1.083$. The gas is net heated everywhere; it is cold because ρ has fallen by a factor 10^2 . The layer is an expansion, not a radiative collapse.

- **The metals-off solution collapses further with no fine-structure cooling at all:** 115 K at $r = 1.033$, where its total cooling is $3.8 \times 10^{-12} \text{ erg cm}^{-3} \text{ s}^{-1}$.

The energy budget makes the same point directly: the metals-on cooling is 36% of the local photoheating at $r = 1.01$ and 2–16% above $r = 1.02$, and heat–cool is balanced by the expansion term $\rho v[d\varepsilon/dr + p d(1/\rho)/dr]$ to within a few percent. The one place radiation dominates is the metals-off base: there H_3^+ carries 100% of the cooling, the entropy *falls* by a factor 3 between the base and $r = 1.02$, and the layer really is radiating itself down.

The closure. Base IR field: True (default False) gives both families of infrared coolant the field they sit in. The lower atmosphere is taken to be black at their wavelengths and to radiate $B_\nu(T_0)$ over the sky fraction $f = 1 - \sqrt{1 - (R_p/r)^2}$, i.e. the whole lower hemisphere at the base and the usual dilution far away. For the eight fine-structure lines the incident field enters *inside* the ground-term statistical equilibrium, as the photon occupation number

$$\bar{n} = \frac{\beta_1(\tau_{\text{dn}})f}{\exp(E_{ul}/kT_0) - 1}, \quad (8)$$

with radiative rates $\beta A(1 + \bar{n})$ down and $\beta A(g_u/g_l)\bar{n}$ up, so the returned power is the *net* one, emission minus absorption. The escape probability becomes two-sided, $\beta = \beta_1(\tau_{\text{up}}) + \beta_1(\tau_{\text{dn}})$ with β_1 the single-face Hollenbach & McKee (1979) / de Jong et al. (1980) form; with the field off it reduces to $2\beta_1(\tau_{\text{up}})$, the previous value, bit for bit. Each line then stops cooling at its own radiative equilibrium temperature, 575.8 K (C I 609 μm) to 641.7 K ([O I] 44 μm) for $T_0 = 1140$ K and half-sky coverage, and heats below it.

For H_3^+ there is no line list (Miller et al. (2013) fit the *total* emission), so the absorption is taken from that fit at the radiating temperature, which is what Kirchhoff’s law makes of an emission integral: $\Lambda_{\text{net}} = n(\text{H}_3^+) 4\pi s(T, n_{\text{H}_2}) [E(T) - W E(T_0)]$. This is the same net-exchange form the H_2 , H_2O and CO channels use (§10.), and it is bounded at every temperature. Its radiative equilibrium temperature is 975.4 K at the base for $T_0 = 1140$ K. Between 2026-08-13 and 2026-08-31 the exchange was instead closed on one effective band, the ν_2 fundamental at 2521.3 cm^{-1} ($E/k = 3627.5$ K), as $\Lambda_{\text{net}} = \Lambda_{\text{emit}}(T) [1 + \bar{n} - \bar{n} e^{E/kT}]$, with equilibrium temperatures of 936.1 K and then 941.2 K. Pairing a total emission fit with the Boltzmann factor of one of its transitions diverges below about 150 K and is wrong by a factor 1.5–1.9 over 400–1000 K; §112 of docs/Update_EXHALE.pdf measures it and records what moved. The approximations and where they break are written at `h3p_net_cooling_rate` (src/modules/lower_atmosphere/h3p_cooling.f90) and at `fine_structure_line_transfer` (src/modules/radiation/Cool_coeff.f90). Fe II (a precomputed statistical-equilibrium table), the coronal remainders and every permitted line keep the optically thin, no-incident-field limit; the estimated heating the remainders would add is $\sim 0.4\%$ of the losses of this layer.

What it does. Both runs restart the converged `Resid tol: 1.0e-5` solutions of §7. with the switch on, and both return `info = 0`.

hot-Uranus gate	metals off		metals on	
	field off	field on	field off	field on
residual norm at exit	6.7×10^{-7}	1.3×10^{-6}	1.4×10^{-6}	9.9×10^{-6}
T at $r = 1.005$	626.8 K	910.0 K	1099.0 K	1098.9 K
T at $r = 1.02$	299.6 K	859.4 K	928.9 K	928.9 K
T at $r = 1.03$	140.2 K	840.2 K	820.0 K	820.1 K
T at $r = 1.05$	476.8 K	753.0 K	573.0 K	573.4 K
coldest cell	115.1 K @ 1.0334	229.7 K @ 1.0947	187.9 K @ 1.0831	189.1 K @ 1.0831
$H_2 \rightarrow H$ front	1.03582	1.08833	1.07976	1.07979
H_3^+ peak [cm^{-3}]	3.75×10^5	3.19×10^5	1.01×10^3	1.00×10^3
$\log_{10} \dot{M}$, spherical	10.539	10.617	10.596	10.596
mass-flux spread, $r > 1.2$	2.4×10^{-3}	2.4×10^{-3}	2.4×10^{-3}	2.4×10^{-3}

Stale, and not re-measurable from here: the field-on columns and the predicted floors below were computed with the single-band H_3^+ closure, which was replaced on 2026-08-31 (§112 of docs/Update_EXHALE.pdf). The converged solutions they restart from no longer exist as run directories, so the table is left as the record of what that closure gave. The same four-way comparison re-measured with the current closure, at the 12000-step protocol of the mol_ir_bands regression case, is in §112.

The metals-off layer stops collapsing and settles *on* the H_3^+ radiative equilibrium curve: the predicted floor with the local sky fraction is 911 K at $r = 1.005$, 886 K at 1.02 and 874 K at 1.03, and the solution sits at 910, 859 and 840 K, just below, by the margin the expansion takes out. Its H_3^+ channel becomes a net *heating* term ($-1.1 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$ at $r = 1.02$) instead of the -1.6×10^{-10} of cooling it had. The front moves out by $0.053 R_p$ and $\dot{M}$ by $+0.078 \text{ dex}$. It is a fixed point of the procedure like the solutions above. The metals-on layer does **not** move (0.1–1.2 K), for the reason measured above: its temperature is set by the expansion, and the fine-structure lines it does carry are 5–20% of the local heating. They now correctly turn into a heating term above $r \approx 1.06$, which is physically right and numerically almost invisible.

What is still open. The metals-on molecular layer, and the outer part of the metals-off one past the $H_2 \rightarrow H$ front (229.7 K at $r = 1.095$), remain far below any radiative equilibrium temperature, and this closure cannot reach them: it only gives the *existing* infrared coolants their incident field, and above the front there are no molecular coolants left. Holding that gas requires a continuum infrared coupling to the deep atmosphere (H_2 collision-induced absorption and the $H_2O/CH_4/CO$ bands), which the model does not have: the whole radiative budget between the base and the front is line channels with a radiative time of $3 \times 10^8 \text{ s}$ against a flow time of $5 \times 10^6 \text{ s}$. Open item (G) is therefore narrowed, not closed.

9. H_2 photodissociation in the Lyman–Werner bands (Stellar LW flux)

The Tier-2 network inherited from Koskinen et al. (2022, Table 1) has no photodissociation of neutral H_2 : its photo-rates start at the 15.4 eV photoionization edge, so below that the only H_2 losses are thermal (R12), electron impact

(R14) and ion chemistry. Their own note says the omission is deliberate and that adding it (Backx et al. 1976 cross section, dissociation probability 0.125) moved their $\dot{M}$ by $\leq 1.4\times$. The consequence for us is recorded: the network wants a base more molecular than any photochemical code gives, and pinning q_{H_2} at the base (Route 1) patches over the missing physics rather than supplying it. This section supplies it.

The band is not in the code's own radiation field. Checked in the source, not assumed. `set_energy_vectors` builds the photon grid from 13.6 eV up; it extends below the H I edge only when the He 2^3S metastable (4.8 eV) or a low-IP metal is active, and then only to that species' threshold, and `read_sed` selects SED rows by the same `e_low`. So for a numerical SED the 11.2–13.6 eV band is simply not read, and for a power-law SED (every Tier-2 gate case) anything below 13.6 eV is the XUV power law extrapolated downward, which has no relation to a star's FUV. Even where the grid does reach into the band, its opacity is continuum photoionization, whereas Lyman–Werner absorption is a forest of saturated lines whose attenuation is nothing like $e^{-\tau_{\text{continuum}}}$. **The band flux is therefore a separate user input, and the transfer is done separately:**

Stellar LW flux [erg/cm2/s]: 343.0 (912–1110 Å, integrated, at the planet)

Default 0 = off. With `Molecular_chemistry` off the key is inert and says so.

Rate. Draine & Bertoldi (1996), ApJ 468, 269 (DB96), published version. Their band is 912–1110 Å: 912 Å is the H Lyman edge, and longward of 1110 Å the H_2 absorptions out of $v = 0$ are negligibly weak (their footnote 4). They characterize a field by the photon flux in that band, $F \equiv c n_{\text{phot}}$ (their eq. 21), and tabulate it with the unshielded dissociation rate. For the flat- F_λ spectrum ($u_\nu \propto \nu^{-2}$, their eq. 24) at $\chi = 1$: $F = 1.208 \times 10^7$ photons $\text{cm}^{-2} \text{s}^{-1}$ (Table 1), $\zeta_{\text{pump}} = 3.09 \times 10^{-10} \text{s}^{-1}$ with $\langle p_{\text{diss}} \rangle = 0.135$ (Table 2), hence $\zeta_{\text{diss}}(0) = 4.17 \times 10^{-11} \text{s}^{-1}$ (Fig. 7 caption). The dissociation rate per band photon is then an effective cross section

$$\sigma_{\text{LW}} = \frac{4.17 \times 10^{-11}}{1.208 \times 10^7} = 3.452 \times 10^{-18} \text{ cm}^2, \quad (9)$$

and with the mean photon energy of a flat- F_λ band, $\langle h\nu \rangle = 2hc/(912 + 1110 \text{ Å}) = 12.2635 \text{ eV}$,

$$k_{\text{LW,thin}} = 1.757 \times 10^{-7} F_{\text{LW}} \quad [\text{s}^{-1}, F_{\text{LW}} \text{ in erg cm}^{-2} \text{s}^{-1}]. \quad (10)$$

Reproducing DB96's own Table 1 numerically from their eqs. (22)–(24) confirms the flux convention (Habing 1.2220×10^7 against their 1.222×10^7 ; ν^{-2} 1.2084 against 1.208; Draine 1978 1.2313 against 1.232, all $\times 10^7$) and the closure of the calibration (the formula returns $4.17 \times 10^{-11} \text{s}^{-1}$ for the ν^{-2} field at $\chi = 1$, their value to 0.04%).

The approximation and its range. σ_{LW} depends on the shape of the spectrum *within* the band, because the Lyman/Werner lines sample it unevenly. The same arithmetic for the much softer Draine 1978 field ($\zeta_{\text{pump}} = 2.78 \times 10^{-10}$, $\langle p_{\text{diss}} \rangle = 0.119$, $F = 1.232 \times 10^7$ at $\chi = 1$) gives $2.685 \times 10^{-18} \text{ cm}^2$, 22% lower, and the formula above would overpredict that field's rate by 26%. The two spectra bracket color temperatures $1.3\text{--}2.9 \times 10^4 \text{ K}$, and DB96 note that PDR properties are insensitive to the spectrum for $T_{\text{color}} \gtrsim 10^4 \text{ K}$. **Take the adopted value as good to $\pm 25\%$ for a stellar FUV band that is not strongly tilted;** a band dominated by a single emission line at one end is outside it.

Self-shielding. DB96 eq. (37), their fit to the full multiline calculation *including line overlap*,

$$f_{\text{shield}}(N_{\text{H}_2}) = \frac{0.965}{(1+x/b_5)^2} + \frac{0.035}{(1+x)^{1/2}} \exp[-8.5 \times 10^{-4}(1+x)^{1/2}], \quad x = \frac{N_{\text{H}_2}}{5 \times 10^{14} \text{ cm}^{-2}}, \quad (11)$$

with $b_5 = b/10^5 \text{ cm s}^{-1}$, and their eq. (40), $\zeta_{\text{diss}} = f_{\text{shield}} e^{-\tau_{\text{dust}}} \zeta_{\text{diss}}(0)$. It reproduces the exact self-shielding function over $10^{14} < N_{\text{H}_2} < 3 \times 10^{21} \text{ cm}^{-2}$ (their Figs. 1–5).

The rate no longer uses that f_{shield} . DB96’s fit is calibrated for cold gas in which only the lowest rotational states of H_2 are populated; Richings, Schaye & Oppenheimer (2014), MNRAS 442, 2780, compared it against a level-resolved CLOUDY calculation and found that it overestimates the shielding factor by a factor ~ 3 . Our molecular layer runs at 945–1307 K, where the overestimate measured with our own columns is 2.8–5.4 $\times$. The photodissociation rate therefore uses their temperature-dependent eqs. (3.12)–(3.15),

$$S_{\text{self}}(N_{\text{H}_2}, T) = \frac{1-w(T)}{(1+x/b_5)^{\alpha(T)}} e^{-5 \times 10^{-7}(1+x)} + \frac{w(T)}{(1+x)^{1/2}} e^{-8.5 \times 10^{-4}(1+x)^{1/2}}, \quad x = \frac{N_{\text{H}_2}}{N_{\text{crit}}(T)}, \quad (12)$$

with $w(T)$, $\alpha(T)$ and $N_{\text{crit}}(T)$ their eqs. (3.13)–(3.15) and the same thermal Doppler parameter: their fit was made for purely thermal broadening, and their Fig. B1 caption states that the agreement with CLOUDY is poorer once turbulence is included, so no turbulent term is added. DB96 eq. (37) is kept, and is still used, for the *band equivalent width* of the shared 912–1110 Å beam: their eq. (39) is the closed-form integral of eq. (37) and of nothing else, and Richings publishes no corresponding equivalent width. `lyman_werner.f90` §2 sets out why the two coexist and which quantity each owns.

b is the H_2 Doppler parameter, taken thermal, $b = (2kT/m_{\text{H}_2})^{1/2}$: 3.07 km s $^{-1}$ at 1140 K, next to the 3 km s $^{-1}$ of DB96’s own figures. N_{H_2} is the star-ward column, built by the same `calc_column_dens_one` radial integration and `opa_pf` weighting as every other absorber column, from the incoming (pre-solve) H_2 density.

Three things are deliberately *not* attenuating the band, all noted at the code: **dust** (EXHALE’s metals are atomic and trace, so there are no grains and the $e^{-\tau_{\text{dust}}}$ of eq. 40 is identically 1); the **trace-metal continuum** (the neutral low-IP metals do photoionize inside the band, but at solar abundance and $\sigma \sim 10^{-18} \text{ cm}^2$ their optical depth is $\sim 4 \times 10^{-23} N_{\text{H}}$, i.e. $\lesssim 0.05$ at a base column where f_{shield} is already below 10^{-4}); and the **H Lyman-series lines** (Ly β 1025.7, Ly γ 972.5, ... all lie in the band), which DB96 include in the equivalent width their fit was built on and state, §4.2, “We will see below that absorption by the H Lyman lines has only a small effect on the H_2 pumping rates”. The last is the least controlled of the three, because a wind has a much larger $N_{\text{HI}}/N_{\text{H}_2}$ than the PDRs DB96 fitted; it can only reduce the rate.

Chemistry and heating. $\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$ enters the H_2 balance row of the coupled system next to the H_2 photoionization (`mol_heh_rows`, `System_HeH_mol`), so it is solved together with everything else rather than in a side loop. Both products are neutral H, which the H-nucleus closure supplies automatically; no other row changes. The fragments carry kinetic energy: Black & Dalgarno (1977), ApJS 34, 405, p. 418, “Fluorescent dissociation of H_2 gives rise to a pair of energetic hydrogen atoms ... for a typical ultraviolet radiation field, the yield is about 0.4 eV per atom pair, but it varies slightly with depth.” We adopt 0.4 eV per dissociation, held constant with depth, added to heat in `ioniz_eq` next to the Penning heating terms. **The 4.48 eV bond energy is paid by the absorbed photon, not by the gas, and is not a sink of this channel** (a thermal dissociation-energy sink for R12/R14 remains a separate open item).

output/Lyman_Werner.txt is written whenever a molecular run carries a band flux: r , T , x_{H_2} , n_{H_2} , N_{H_2} , f_{shield} , k_{LW} and the photodissociation heating.

The band flux for a planet. The code's own SED cannot supply it, so it is integrated externally. The VULCAN stellar spectra in the tree (VULCAN/atm/stellar_flux/, flux at the *stellar surface*, $\text{erg cm}^{-2} \text{s}^{-1} \text{nm}^{-1}$) integrated over 91.2–111.0 nm and diluted by $(R_*/a)^2$:

star / planet	surface band flux	R_* , a	F_{LW} at the planet
Sun, Gueymard_solar.txt, at 1 au	2.74×10^4	$1 R_\odot$, 1 au	0.592
HD 209458 b (solar spectrum as proxy)	2.74×10^4	$1.155 R_\odot$, 0.048 au	343
HD 189733 b, sflux-HD189_Moses11.txt	4.23×10^4	$0.805 R_\odot$, 0.03142 au	600
HD 189733 b, sflux-HD189_B2020.txt	1.29×10^5	$0.805 R_\odot$, 0.03142 au	1.83×10^3

($\text{erg cm}^{-2} \text{s}^{-1}$ throughout; the corresponding $k_{\text{LW,thin}}$ are 1.04×10^{-7} , 6.03×10^{-5} , 1.05×10^{-4} and $3.21 \times 10^{-4} \text{s}^{-1}$.) The VULCAN_run_*/atm/stellar_flux/ directories of the workspace are empty (casualties of the 2026-08-09 loss), so no HD 209458 spectrum exists in the tree and the solar spectrum stands in for it; HD 209458 is a G0 star of comparable activity, but **$343 \text{ erg cm}^{-2} \text{s}^{-1}$ is a proxy, not a measurement**. The calibration check on the same file is the solar $\text{Ly}\alpha$ irradiance it returns at 1 au, $6.12 \text{ erg cm}^{-2} \text{s}^{-1}$ against an observed 6–8. The factor 3 between the two HD 189733 spectra is a fair measure of how uncertain a real stellar FUV band flux is, and it dwarfs the $\pm 25\%$ of the band-shape approximation.

What it does: hot-Uranus gate, A/B on the key alone (metals off). Both runs restart the mol_base_handoff configuration ($q_{\text{H}_2}^{\text{base}} = 0.75$, $p_{\text{base}} = 10^{-5} \text{ bar}$) with the three convergence keys of §7. and Base IR field: True, and differ only in Stellar LW flux. Both return info = 0.

	LW off	LW on (343)
residual norm at exit	8.15×10^{-6}	4.86×10^{-6}
$\text{H}_2 \rightarrow \text{H}$ front ($x_{\text{H}_2} = 0.5$)	1.08806	1.08622
x_{H_2} at the base	0.99921	0.99883
q_{H_2} at the base	0.8618	0.8612
x_{H_2} at $r = 1.05$	0.9777	0.9629
x_{H_2} at $r = 1.10$	2.54×10^{-2}	1.51×10^{-4}
T at $r = 1.005/1.05/1.10$	901.5 / 752.3 / 255.6 K	886.6 / 751.8 / 249.9 K
$\log_{10} \dot{M}$, spherical	10.6196	10.6248

The self-shielding works as designed and is visible in the dump: $N_{\text{H}_2} = 4.32 \times 10^{21} \text{ cm}^{-2}$ and $f_{\text{shield}} = 9.79 \times 10^{-7}$ at the base, 2.85×10^{-4} at the front ($r = 1.087$), 0.467 at $r = 1.10$ and exactly 1 above $r = 1.15$, where k_{LW} recovers the unshielded $6.026 \times 10^{-5} \text{s}^{-1}$ computed above. The band is unattenuated in the wind and suppressed by 10^6 at the base.

Route 2 verdict: the physics was missing, but it does not close the gap. The Route 2 criterion of base_composition_handoff_plan.md §5 was whether the network's own base composition moves toward the photochemical partition once the missing dissociation is supplied, without pinning it. **It does not**, and the measurement says

why. The base q_{H_2} moves from 0.8618 to 0.8612 against a photochemical target of 0.75: 0.5% of a gap of 0.11. Yet Lyman–Werner *is* the largest single H_2 loss term at the base after the change: $5.90 \times 10^{-11} \text{ s}^{-1}$ against 2.52×10^{-11} for $\text{H}^+ + \text{H}_2$ (R10 + R13), 5.56×10^{-12} for $\text{He}^+ + \text{H}_2$ and 3.87×10^{-13} for thermal dissociation. The two facts are consistent because the base partition is a formation–destruction balance in which the three-body reaction R15 ($\text{H} + \text{H} + \text{M}$) sets $n_{\text{H}_2} \propto n_{\text{H}}^2$, so

$$\frac{n_{\text{H}}}{n_{\text{H}_2}} \propto (\text{total } \text{H}_2 \text{ destruction rate})^{1/2}, \quad (13)$$

and the square root is brutal: the measured $2.3 \times$ rise in the destruction rate raises the atomic fraction by only $1.5 \times$, from 7.9×10^{-4} to 1.2×10^{-3} . Reaching $q_{\text{H}_2} = 0.75$ means $x_{\text{H}_2} = 0.925$, i.e. an atomic fraction $64 \times$ larger, i.e. a destruction rate **4100** $\times$ larger: about $3.7 \times 10^{-7} \text{ s}^{-1}$. The unshielded band would supply that many times over ($6.0 \times 10^{-5} \text{ s}^{-1}$); it is stopped by the column. The base of this planet sits under $N_{\text{H}_2} = 4.3 \times 10^{21} \text{ cm}^{-2}$, where $f_{\text{shield}} = 9.8 \times 10^{-7}$, and f_{shield} reaches the required $\sim 6 \times 10^{-3}$ only near $N_{\text{H}_2} \sim 10^{17} - 10^{18} \text{ cm}^{-2}$, four orders of magnitude shallower.

So Lyman–Werner photodissociation cannot be the reason photochemical codes find more atomic H at 1 μbar .

That reading is not new physics: §2 of this memo already records that photochemistry (OH-catalyzed H_2 destruction) dissociates H_2 even at 500–1000 K where thermal equilibrium would not. The catalytic cycles run on O, OH and H_2O , and EXHALE’s molecular network is $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ with atomic metals; it has nowhere to put them. Closing the base gap needs those species, not a stronger radiation field. Route 1 (pinning $q_{\text{H}_2_base}$) therefore remains the only way EXHALE can carry the photochemical base partition today, and it is now a documented modeling choice rather than a patch over a missing term.

What Lyman–Werner *does* change is the top of the molecular layer, where the column has fallen enough for the band to bite: at $r = 1.10$ the H_2 fraction drops by a factor 168, the front moves inward by $0.0018 R_p$, and $\dot{M}$ rises by 0.005 dex. Small, and in the direction and of the order Koskinen et al. (2022) reported for their own sensitivity test.

The same A/B on HD 209458 b. The examples/15_molecular configuration (metals off, Base IR field: True added, both runs restarted from its converged solution) reproduces the pattern on a real planet. Here the JFNK finish would not reach `Resid tol: 1.0e-5` with the band on: it stalled three times at $\|R\| = 1.5 - 6.8 \times 10^{-5}$ on the base-adjacent momentum row, open item (A), so *both* members of the pair were run at `Resid tol: 2.0e-5`, where both `return info = 0` and both mass fluxes are flat to 4.1×10^{-3} over $r > 1.2$.

	LW off	LW on (343)
residual norm at exit	1.77×10^{-5}	1.96×10^{-5}
$\text{H}_2 \rightarrow \text{H}$ front ($x_{\text{H}_2} = 0.5$)	1.01578	1.01290
x_{H_2} at the base	0.99628	0.99605
q_{H_2} at the base	0.8512	0.8509
x_{H_2} at $r = 1.02$	2.37×10^{-4}	4.22×10^{-6}
H_3^+ peak [cm^{-3}]	4.81×10^5 @ 1.0010	5.46×10^5 @ 1.0010
T at $r = 1.005/1.02$	937.7 / 986.3 K	792.7 / 1275.8 K
$\log_{10} \dot{M}$, spherical	10.5711	10.5600

The base is again untouched ($N_{\text{H}_2} = 2.6 \times 10^{21} \text{ cm}^{-2}$, $f_{\text{shield}} = 2.1 \times 10^{-6}$), the front moves in by $0.0029 R_p$, the H_2

fraction just above it falls by $56\times$, and $\dot{M}$ moves by -0.011 dex. The chemical-equilibrium fit gives $q_{\text{H}_2} = 0.831$ at this base and photochemistry would give less still; the network sits at 0.851 with the band on as without it.

10. The molecular infrared bands (Molecular IR bands, 2026-08-31)

The Base IR field closure of §8. gave the infrared coolants the code already had (the eight ground-term fine-structure lines and the H_3^+ bands) the field they sit in, and stopped there. Its own closing paragraph named what was left: past the $\text{H}_2 \rightarrow \text{H}$ front there are no molecular coolants in the model at all, and holding that gas needs the coolants a real H_2 atmosphere carries. This section supplies three of them, and addresses open item (G).

Which coolants, and why these three. H_2 quadrupole and magnetic dipole lines (Roueff et al. 2019, table 2, VizieR J/A+A/630/A58). H_2 is the bulk gas: its transition probabilities are 10^{-10} – 10^{-6} s^{-1} , but it outnumbers every other molecule by 10^3 – 10^4 , and the two cancel. At $n(\text{H}_2) = 5 \times 10^{13} \text{ cm}^{-3}$ and 1000 K it radiates $1.4 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$, four times the local photoheating (the pure rotational part alone is 4.8×10^{-8} ; above about 600 K the vibrational bands carry most of it). It is the only one of the three that exists in a Tier-2 run without the oxygen option. H_2O (HITEMP 2010) is the strongest infrared coolant of a solar-composition H_2 atmosphere below ~ 2000 K, and the one the oxygen option switches the [O I] fine-structure lines off in favour of. CO (HITEMP 2019) holds essentially all the carbon at the base; its $4.7 \mu\text{m}$ fundamental carries four to five orders of magnitude more than its pure rotational lines above 300 K.

H_2 – H_2 and H_2 –He collision-induced absorption is deliberately *not* a channel, and that is a measurement rather than a scope decision. It scales as n^2 , and on the CIA tables shipped with Photochem its optical depth over a pressure scale height at the $1 \mu\text{bar}$ base is 8×10^{-9} (hot Uranus) to 2×10^{-9} (HD 189733 b), with unit optical depth reached only near 0.2–0.4 bar: consistent with Lavvas & Arfaux (2021), who put CIA’s onset at $p > 1$ bar. It is what makes the reservoir *below* the base black in the windows between the bands (the assumption Base IR field already encodes), not a local coolant of the modeled domain.

The closure. For each species the net radiative loss per unit volume is the optically thin, LTE exchange with the diluted field of the lower atmosphere,

$$\Lambda_{\text{net}} = n_X [E_X(T) - W E_X^{\text{abs}}(T; T_0)], \quad (14)$$

with, for a band absorber,

$$E_X(T) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T) d\nu, \quad E_X^{\text{abs}}(T; T_0) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T_0) d\nu, \quad (15)$$

and, for the H_2 line sum, the same quantity assembled transition by transition with the full two-level net factor $1 + \bar{n} - \bar{n} e^{\Delta E/T}$. W is the dilution the H_3^+ closure uses, $\frac{1}{2}[1 - \sqrt{1 - (R_p/r)^2}]$, and it is zero when Base IR field is off.

Keeping the stimulated-emission term is not decoration: it is what makes the bracket vanish identically at $T = T_0$, $W = 1$, gas buried in a blackbody at its own temperature neither cools nor heats. For the band absorbers that property is automatic, because the HITRAN line intensities the cross sections are built from already carry the $1 - e^{-h\nu/kT}$ factor, so σ_b is the coefficient that pairs with B_ν . For a line sum it has to be written down. The H_3^+ closure of §8. had dropped it; it

was put back the same day, and because $E_k(\nu_2)/T_0$ is large that cost only 2.2% of the emission and 5 K of equilibrium temperature there ($936 \rightarrow 941$ K at $T_0 = 1140$ K). On a far-infrared band the same omission would be an order-unity error. The single-band H_3^+ form the 941 K belongs to was itself replaced the same day (a total emission fit cannot be paired with the Boltzmann factor of one transition at all temperatures), and the H_3^+ column below is the replacement's (§112 of docs/Update_EXHALE.pdf).

Each channel therefore has its own radiative equilibrium temperature, and it is the fixed point rather than the magnitude that answers item (G):

T_0	H_2O	CO	H_2	(H_3^+)
900 K	705 K	752 K	783 K	780.9 K
1140 K	889 K	918 K	984 K	975.4 K
1183 K	922 K	947 K	1019 K	1010.7 K
1450 K	1120 K	1120 K	1226 K	1231.6 K
1800 K	1369 K	1338 K	1486 K	1517.6 K
2358 K	1749 K	1668 K	1883 K	1938.6 K

(half-sky coverage, $W = \frac{1}{2}$). How much rests on that assumption is measurable: for H_2O at $T_0 = 1183$ K the equilibrium temperature is $726.5 / 922.1 / 1065.0 / 1183.0$ K at $W = 0.25 / 0.5 / 0.75 / 1$, and CO $784.7 / 946.5 / 1073.1 / 1183.0$ K, at $W = 1$ both are exactly T_0 , the closure's fixed point recovered numerically. Because W multiplies only the absorption, these temperatures are insensitive to first order to any grey escape probability applied to both terms, which is why the optically thin limit is enough to fix the *temperature* even where a band is marginally thick.

Optical depth, and why the thin limit is used. At a $1 \mu\text{bar}$ base with solar oxygen the band-mean optical depth of the H_2O rotational bands over a scale height is a few times 10^{-2} ; the strongest g -points inside a band reach order unity. The H_2 lines are thinner still: the line-centre opacity of 0–0 S(1) at $17 \mu\text{m}$ gives $\tau \approx 3 \times 10^{-4}$ over a scale height at $n(\text{H}_2) = 5 \times 10^{13} \text{ cm}^{-3}$ and 1000 K. No escape probability is applied, and the Planck-mean optical depths of the H_2O and CO columns to the top of the domain are written to output/Cooling_breakdown.txt so the assumption is measured rather than asserted. output/FUV_bands.txt carries the same statement as a conservation check: the column-integrated absorbed power of the three bands against the incident infrared flux $W\sigma T_0^4$, with a warning if it is exceeded.

Where the numbers come from, and three independent checks. The band-mean cross sections are $\sum_g w_g k_g$ of the correlated- k tables, which is exactly the bin-mean cross section. Line strengths do not depend on the broadening, only the line shape does, so this mean is pressure independent to better than 2% from 10^{-6} to 1 bar (measured), and the whole EXHALE domain uses the 10^{-6} bar slice, which is also the lowest tabulated pressure. The tables run 50–2000 K and are clamped outside; the H_2 line sum has no such limit. Three checks, against sources that share none of that line processing:

- the Planck-mean H_2O cross section agrees with the independent table aiolos/inputdata/1H2-160-T400.aiopa to within 11% over 600–2000 K (ratio 0.889–1.111 on the 14 shared grid temperatures); be-

low 600 K that table is not monotonic in T and the comparison is not meaningful, the low-temperature end being covered by the Neufeld & Kaufman comparison instead;

- restricted to $\lambda > 9 \mu\text{m}$, i.e. to the pure rotational band, the H_2O emission agrees with the optically thin LTE rate of Neufeld & Kaufman (1993) table 2 to a factor 0.70 (100 K), 0.81 (200 K), 0.90 (400 K), 1.02 (1000 K); the same cut on CO against their table 3 gives 0.81–0.93 over 300–2000 K. At 100 K the CO ratio falls to 0.10, most likely the eight-point g -quadrature resolving the linear bin mean poorly for CO’s sparse rotational ladder rather than the coarse long-wavelength binning (97% of the 100 K emission lands in the eighteen 9–250 μm bins); CO radiates $1.6 \times 10^{-20} \text{ erg s}^{-1}$ per molecule there against 1.8×10^{-15} for H_2O , so it does not matter;
- the H_2 line sum agrees with the LTE rates of Hollenbach & McKee (1979) eqs. (6.37) and (6.38) to 1–11% over 200–2000 K.

Two numbers from those comparisons are the reason the whole band system is integrated instead of a published *rotational* cooling function being adopted: above 300 K CO’s total emission is four to five orders of magnitude above its rotational part, and H_2O ’s is $3.2\times$ its rotational part at 1000 K. Neufeld & Kaufman’s tables also stop at an optical depth parameter of $10^{19} \text{ cm}^{-2} (\text{km/s})^{-1}$, a ceiling set by hydrostatic equilibrium in a self-gravitating cloud, which does not apply to a planetary atmosphere and which a base column exceeds.

What it does. Four converged solutions of the hot-Uranus gate (data_g2 restarted with Load IC? True, Solver: Newton 5.0e-2, Resid tol: 1.0e-5, Max steps: 150000), all with Base IR field: True, so each pair differs only in Molecular IR bands. This configuration has no oxygen chemistry, so the only new channel is the H_2 line sum. The bands-off members reproduce the field-on solutions of §8. to 1–3 K (metals on) and 5–30 K (metals off, the difference being the H_3^+ stimulated-emission fix).

	metals OFF		metals ON	
	off	on	off	on
JFNK iterations	89	21	25	26
residual norm at exit	8.77e-6	8.39e-6	2.00e-6	2.82e-6
$T(r = 1.005)$	940.5 K	912.1 K	1097.0 K	1044.1 K
$T(r = 1.02)$	878.6 K	857.6 K	926.1 K	887.0 K
$T(r = 1.05)$	750.3 K	721.9 K	572.2 K	588.7 K
$T(r = 1.08)$	420.9 K	345.4 K	215.1 K	236.1 K
coldest cell below 1.3	245.6 K @ 1.0972	225.4 K @ 1.0923	190.4 K @ 1.0842	198.2 K @ 1.0853
$\text{H}_2 \rightarrow \text{H}$ front	1.08817	1.08532	1.07916	1.07970
$\log_{10} \dot{M}$ reported	10.32	10.31	10.29	10.30
mass-flux spread, $r > 1.2$	2.39e-3	2.38e-3	2.38e-3	2.38e-3

The solution’s own zero crossing lands on the predicted equilibrium temperature: in the metals-on solution the H_2 channel changes sign at $r = 1.0122$, where the gas is at 951.7 K and the sky dilution is $W = 0.423$, and the closure evaluated at that dilution gives 951.5 K; the second crossing ($r = 1.1239$, $T = 850.6 \text{ K}$, $W = 0.272$) is predicted at 872.7 K. Nothing was fitted.

The H_2 channel is *first order and does not repair the collapse*, and those are not in tension. In the metals-on solution H_2 carries 56% of the whole radiative rate at $r = 1.002\text{--}1.005$ (cooling there, since $T = 1044\text{--}1124$ K is above its radiative equilibrium temperature), and essentially all of the net at $r = 1.02\text{--}1.05$, heating. The layer's radiative time falls with it: $t_{\text{rad}}/t_{\text{flow}}$ at $r = 1.05$ goes from **8.22 to 0.263**. But the coldest cell sits *at or beyond* the $\text{H}_2 \rightarrow \text{H}$ front in every run, where there is no H_2 to absorb anything and $t_{\text{rad}}/t_{\text{flow}}$ is 0.7–1.9; that gas is expansion-dominated and no molecular coolant can reach it.

With the oxygen chemistry on, where H_2O and CO exist, the magnitudes differ by five orders: on HD 189733 b the two bands carry 4.8×10^{-5} and 2.7×10^{-5} $\text{erg cm}^{-3} \text{ s}^{-1}$ at $r = 1.005$ against a total of 7.5×10^{-5} , and a 12000-step A/B moves the base from 760.2 to 829.1 K and x_{H_2} from 0.330 to 0.483, both toward the photochemical reference (864 K, 0.910). Neither of those runs is converged; docs/Update_EXHALE.md §110 carries the full statement and its caveats.

What is assumed. LTE level populations (the rotational levels thermalize far below the densities of this layer; $n_{\text{crit}}(\text{H}_2 v=1)$ is $10^9\text{--}10^{11} \text{ cm}^{-3}$ against $10^{13}\text{--}10^{14} \text{ cm}^{-3}$ at the base, and the vibrational bands are only 0.7% of the H_2 emission at 400 K, 5.7% at 500 K and 18% at 600 K, so the overestimate is confined to the warm, dense part of the layer). The incident field is not attenuated by the intervening molecular column of the domain itself, the same approximation Base IR field makes for H_3^+ . OH is left out: it has no cross-section table here and carries a few percent of the oxygen where the water carries most of it. The atomic advection post-process (T_equation) does not carry these channels, exactly as it does not carry H_3^+ .

References

- [1] Banks, P. M., & Kockarts, G. 1973, *Aeronomy* (New York: Academic Press)
- [2] Black, J. H., & Dalgarno, A. 1977, *ApJS*, 34, 405
- [3] de Jong, T., Boland, W., & Dalgarno, A. 1980, *A&A*, 91, 68
- [4] Draine, B. T., & Bertoldi, F. 1996, *ApJ*, 468, 269
- [5] Guillot, T. 2010, *A&A*, 520, A27
- [6] Hollenbach, D., & McKee, C. F. 1979, *ApJS*, 41, 555
- [7] Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, *ApJ*, 851, 150
- [8] Huang, C., Koskinen, T., Lavvas, P., & Fossati, L. 2023, *ApJ*, 951, 123
- [9] Koskinen, T. T., Harris, M. J., Yelle, R. V., & Lavvas, P. 2013a, *Icarus*, 226, 1678
- [10] Koskinen, T. T., Lavvas, P., Huang, C., et al. 2022, *ApJ*, 929, 52
- [11] Larsson, M., McCall, B. J., & Orel, A. E. 2008, *Chem. Phys. Lett.*, 462, 145
- [12] Richings, A. J., Schaye, J., & Oppenheimer, B. D. 2014, *MNRAS*, 442, 2780
- [13] Roueff, E., Abgrall, H., Czachorowski, P., Pachucki, K., Puchalski, M., & Komasa, J. 2019, *A&A*, 630, A58

- [14] Neufeld, D. A., & Kaufman, M. J. 1993, *ApJ*, 418, 263
- [15] Hollenbach, D., & McKee, C. F. 1979, *ApJS*, 41, 555
- [16] Rothman, L. S., Gordon, I. E., Barber, R. J., et al. 2010, *JQSRT*, 111, 2139
- [17] Li, G., Gordon, I. E., Rothman, L. S., et al. 2015, *ApJS*, 216, 15
- [18] Grimm, S. L., & Heng, K. 2015, *ApJ*, 808, 182
- [19] Wogan, N. F., Catling, D. C., Zahnle, K. J., & Kopparapu, R. K. 2025, *PSJ*, 6, 256
- [20] Lavvas, P., Koskinen, T., & Yelle, R. V. 2014, *ApJ*, 796, 15
- [21] Lavvas, P., Koskinen, T., Steinrueck, M. E., García Muñoz, A., & Showman, A. P. 2019, *ApJ*, 878, 118
- [22] Lavvas, P., & Arfaux, A. 2021, *MNRAS*, 502, 5643
- [23] Malik, M., Grosheintz, L., Mendonça, J. M., et al. 2017, *AJ*, 153, 56
- [24] Miller, S., Stallard, T., Tennyson, J., & Melin, H. 2013, *J. Phys. Chem. A*, 117, 9770
- [25] Moses, J. I., Visscher, C., Fortney, J. J., et al. 2011, *ApJ*, 737, 15
- [26] Neale, L., Miller, S., & Tennyson, J. 1996, *ApJ*, 464, 516
- [27] Salz, M., Czesla, S., Schneider, P. C., & Schmitt, J. H. M. M. 2016, *A&A*, 586, A75
- [28] Storey, P. J., & Hummer, D. G. 1995, *MNRAS*, 272, 41
- [29] Taylor, A. R., Koskinen, T., Argenti, L., et al. 2025, *ApJ*, 989, 68
- [30] Taylor, A. R., Koskinen, T., Huang, C., Arfaux, A., & Lavvas, P. 2026, *ApJ*, 999, 214
- [31] Tsai, S.-M., Lyons, J. R., Grosheintz, L., et al. 2017, *ApJS*, 228, 20
- [32] Tsai, S.-M., Malik, M., Kitzmann, D., et al. 2021, *ApJ*, 923, 264
- [33] Visscher, C., Lodders, K., & Fegley, B. 2006, *ApJ*, 648, 1181
- [34] Voronov, G. S. 1997, *At. Data Nucl. Data Tables*, 65, 1
- [35] Xing, L., Yan, D., & Guo, J. 2023, *ApJ*, 953, 166